

A-MES

AUTOMOTIVE MES · SEOYON E-HWA

VOL 03

Warehouse Management (WH)

Inbound · Inventory · Release · Locations

Document · L3 Detailed Design Specification

Generated · 05-26-2026 · Build 20260526-190315

Reference · Plant A · SAV Detroit, MI | Plant B · GEO Birmingham, AL

CONFIDENTIAL · FOR INTERNAL USE ONLY

TABLE OF CONTENTS

Warehouse Management (WH)

WH_Warehouse	A-MES · VOL.3 · WH · Warehouse Management
WH01_Inbound_Schedule	A-MES · VOL.3 · WH-01 · Inbound Schedule · 상세 설계서
WH02_PDA_Inbound	A-MES · VOL.3 · WH-02 · PDA Inbound Receiving · 상세 설계서
WH03_Inventory_Status	A-MES · VOL.3 · WH-03 · Inventory Status · 상세 설계서
WH04_Location_Map	A-MES · VOL.3 · WH-04 · Location Map · 상세 설계서
WH05_Inventory_Adjust	A-MES · VOL.3 · WH-05 · Inventory Adjustment · 상세 설계서
WH06_Release_Schedule	A-MES · VOL.3 · WH-06 · Release Schedule · 상세 설계서
WH07_PDA_Release	A-MES · VOL.3 · WH-07 · PDA Release Picking · 상세 설계서
WH08_Transaction_History	A-MES · VOL.3 · WH-08 · Transaction History · 상세 설계서

WAREHOUSE MANAGEMENT

창고 관리 — PDA 바코드 · FIFO 출고 · 실시간 재고 · 입출고 이력

Full inbound/outbound warehouse management. **Key features:** SCM PO auto-receive (no PO = scan blocked), PDA barcode/QR scan for both inbound and outbound, real-time inventory by location, automatic min-stock alerts, mandatory FIFO picking (oldest LOT first), and full transaction audit trail. Integrates with PP (stock for MRP) and FG (finished goods inbound).

입고·출고 통합 창고 관리. **핵심 기능:** SCM 발주 자동 수신 (PO 없는 품목 스캔 차단), 입고/출고 모두 PDA 바코드·QR 스캔, 로케이션별 실시간 재고, 최소재고 자동 알림, FIFO 출고 강제 (입고일 오래된 LOT 우선), 입출고 전수 감사 이력. PP(MRP 재고)·FG(완제품 입고) 연계.

SCREENS	PDA COVERAGE	FIFO RULE	QC TRIGGER	MIN STOCK	SCM
8	2 (In · Out)	Mandatory	Auto on Inbound	Alert MD-09/10	PO Auto-Sync

Click any screen card / icon / pill for L3 detailed spec · 각 화면 카드 클릭 시 L3 상세 설계서

WH-01 Inbound Schedule

WH-02 PDA Inbound

WH-03 Inventory Status

WH-04 Location Map

WH-05 Inventory Adjust

WH-06 Release Schedule

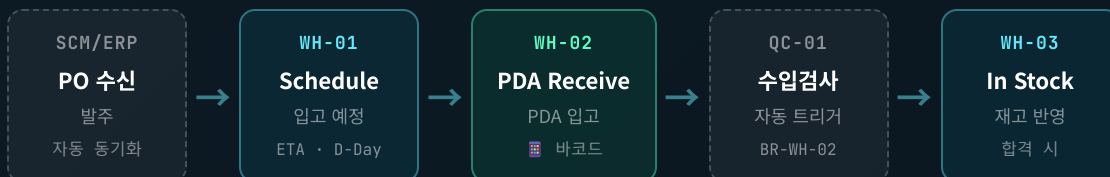
WH-07 PDA Release

WH-08 Transaction History

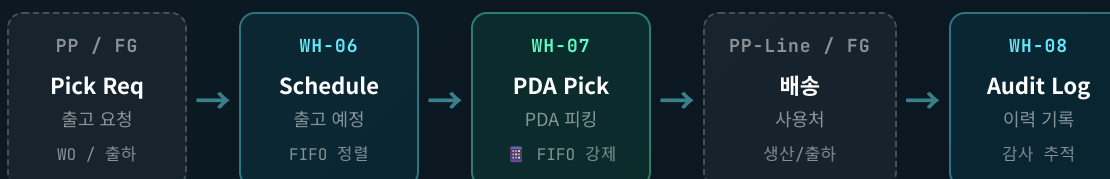
PROCESS FLOW · 화면 흐름

INBOUND & OUTBOUND TRACKS · 입출고 양방향 흐름

INBOUND TRACK · 입고



OUTBOUND TRACK · 출고



WH-04

Map

로케이션 맵
QR Print

WH-05

Adjust

재고 조정
PIN + 사유

WH-03

Status

현황
실시간 PDA 화면  Web 화면  외부 모듈

▼ MODULE KPI · 핵심 지표

TODAY INBOUND

금일 입고 건수

TODAY OUTBOUND

금일 출고 건수

STOCK ACCURACY

 $\geq 99\%$

목표 / 분기 cycle count

FIFO COMPLIANCE

100%

강제 (BR-WH-04)

MIN STOCK ALERT

현 알림 건수

▼ 8 SCREENS · 화면 설계



WH-01

Inbound Schedule

입고 예정 — SCM PO 자동 수신 + ETA 관리

 Web PC

Phase 0 + 1st

 L3 DETAIL →

What you'll see: SCM 발주 자동 수신 목록 + ETA D-Day 자동 계산 + 지연 빨강 강조

[Receive] 버튼 클릭 시 WH-02 PDA 입고 흐름으로 자동 분기

 INPUT

SCM PO Auto-Sync

 OUTPUT

Schedule + WH-02 trigger

 WATCH

Overdue (red)

 UI

Web Table + Filter

EN · DESCRIPTION

Displays all open purchase orders received from the SCM portal and allows warehouse staff to register expected inbound schedules. ETA D-Day is auto-calculated; overdue deliveries are highlighted in red. Clicking [Receive] triggers the WH-02 PDA inbound flow.

KO · 설명

SCM 포탈 수신 발주 기반 입고 예정 등록 화면. 입고 ETA D-Day 자동 계산, 지연 건 빨간색 강조. [Receive] 클릭 시 WH-02 PDA 입고 흐름으로 연계.



Auto PO Sync

SCM 포탈 hourly polling.
신규 PO 자동 수신

ETA D-Day

자동 계산 + 카운트다운.
D-3 노랑, D-0 빨강

Overdue Highlight

납기 초과 행 빨간 배경 + 알림.
SCM 자동 조회 알림

→ WH-02 Trigger

[Receive] 클릭 시 PDA 입고 진입.
선택 행 자동 인계

관련 BR ▶

BR-WH-01



WH-02 · ⚙️ CORE

PDA Inbound Receiving

PDA 입고 처리 — 바코드 스캔 + 위치 지정

PDA

⚙️ Core

L3 DETAIL →



What you'll see: PDA 스캔 화면 (품목 바코드 → 수량 숫자패드 → 선반 QR) 3단계

PO 미등록 품목은 첫 스캔 즉시 차단 + QC 검사 자동 트리거

INPUT

Item Barcode + Qty + Loc
QR

OUTPUT

Stock IN (QC pending)

BLOCK

PO 없는 품목

UI

3-Step PDA Flow

EN · DESCRIPTION

Warehouse staff scans item barcode via PDA, confirms quantity using the numeric keypad, then scans a shelf QR code to assign storage location. System validates barcode against PO. Mismatch is immediately blocked. Confirms stock entry, triggers QC-01 inspection, prints location label.

KO · 설명

창고 담당자가 PDA로 품목 바코드 스캔 → 숫자패드로 수량 입력 → 선반 QR 스캔으로 로케이션 지정. 바코드-PO 불일치 즉시 차단. 입고 확정 + QC-01 검사 자동 트리거 + 위치 라벨 출력.



3-Step Scan

품목 바코드 → 수량 패드 → 위치 QR.
1화면 1액션 원칙



PO Validation

미등록 바코드 즉시 차단.
BR-WH-01



Auto QC Trigger

입고 즉시 QC-01 수입검사 발행.
합격 후 재고 반영



Location Label

선반 QR 라벨 자동 인쇄.
위치 부착용

관련 BR ▶

BR-WH-01

BR-WH-02



WH-03

Inventory Status

재고 현황 — 로케이션별 실시간

Web PC

1st Dev

L3 DETAIL →



What you'll see: 품목/LOT/로케이션별 실시간 재고 그리드 + 최소재고 임박 노랑 강조

행 클릭 시 WH-08 입출고 이력 드릴다운

INPUT

All WH transactions

OUTPUT

Stock view + Drill-down

ALERT

Min Stock (BR-WH-03)

UI

Searchable Web Grid

EN · DESCRIPTION

Real-time inventory query by item, LOT, location, or expiry.
Min-stock items highlighted yellow with alert push to warehouse manager + procurement team. Drill-down per row to WH-08 transaction history. Excel export for cycle count.

KO · 설명

품목·LOT·로케이션·유효기간 실시간 검색. 최소재고 미만 행 노랑 강조 + 창고담당·구매팀 즉시 알림. 행 클릭 시 WH-08 입출고 이력 드릴다운. cycle count용 Excel 출력.



Multi-Search

품목·LOT·위치·기간 동시 검색.
필터 조합



Min Stock Alert

MD-09/10 기준 임박 시 노랑·미만 시 빨강.
+ 즉시 알림



Drill-down

행 클릭 시 WH-08 이력.
전체 트랜잭션 추적



Excel Export

cycle count + 정기 점검용.
7일 보관

관련 BR ▶

BR-WH-03



WH-04

Location Map

로케이션 맵 — 시각화 + QR 출력

Web PC

2nd Dev

L3 DETAIL →



What you'll see: Zone × Row × Column 그리드 시각화 (빈/부분/만재/초과 4색)

셀 클릭 시 위치별 품목 상세, QR 라벨 일괄 출력 지원

INPUT

MD-11 Location Master

OUTPUT

Visual Map + QR Labels

UI

Grid + Color Coded

ACTION

Reorganize (drag)

EN · DESCRIPTION

Visual warehouse layout: Zone → Row → Column → Level hierarchy. Each cell color-coded by capacity (Empty/Partial/Full/Over). QR label batch print for new locations. Drag to reorganize (supervisor PIN). Direct from MD-11 Location Master.

KO · 설명

창고 레이아웃 시각화: 구역 → 행 → 열 → 층 계층. 셀 용량별 4색(빈/부분/만재/초과) 표시. 신규 위치 QR 라벨 일괄 출력. 드래그로 재배치 (라인장 PIN). MD-11 로케이션 마스터 직접 참조.



4-Color Capacity

Empty/Partial/Full/Over 즉시 인식.
한눈 파악



QR Batch Print

선반 라벨 일괄 출력.
Zebra ZPL



Drag Reorganize

레이아웃 변경 (PIN 필요).
감사 기록



MD-11 Sync

로케이션 마스터 실시간 미러.
변경 즉시 반영

관련 BR ▶ 없음 (참조 화면)



WH-05

Inventory Adjustment

재고 조정 — 사유 코드 + 라인장 PIN

Web PC

1st Dev

L3 DETAIL →



What you'll see: cycle count 차이 또는 손실/발견 시 사유 코드 + 수량 조정 입력 폼

모든 조정은 PIN 인증 + 7년 감사 로그 (BR-WH-05)

TRIGGER

Cycle count diff / Event

OUTPUT

Stock +/- + Audit Log

AUTH

Supervisor PIN

UI

Reason Code Form

EN · DESCRIPTION

Manual inventory adjustment with mandatory reason code (Damaged / Lost / Found / Count_Diff / Other), supervisor PIN, and full audit trail. All changes recorded in WH-08 history with operator, approver, before/after qty. Audit log retained 7 years.

KO · 설명

수동 재고 조정 — 사유 코드 필수 (파손/분실/발견/실사차이/기타), 라인장 PIN 인증, 전수 감사 추적. 모든 변경이 WH-08 이력에 작업자·승인자·전후 수량과 함께 기록. 감사 로그 7년 보관.



Reason Code Required

5개 사유 중 1개 강제 선택.
자유 텍스트 200자



Supervisor PIN

라인장 인증 없이 조정 불가.
감사 강화



Before/After Snap

조정 전후 수량 자동 기록.
롤백 가능



7-Year Audit

WH-08에 영구 기록.
감사 요구사항

관련 BR ▶ 참조: BR-WH-03 (Min Stock)

WH-06



Release Schedule

출고 예정 — 픽 리스트 + FIFO 자동 정렬

[Web PC](#)[1st Dev](#)[L3 DETAIL →](#)

What you'll see: PP WO 자재 요청 + FG 출하 요청 통합 목록 + FIFO 자동 추천 위치

[Pick] 버튼 클릭 시 WH-07 PDA 피킹 화면으로 자동 분기

INPUT

PP WO / FG Shipment

OUTPUT

Pick List + WH-07 trigger

SORT

FIFO Auto

UI

Web Table + Filter

EN · DESCRIPTION

Consolidates outbound requests from PP (material to production line) and FG (finished goods shipment). System auto-suggests pick locations following FIFO rule (oldest LOT first). [Pick] button transitions to WH-07 PDA picking flow with selected items.

KO · 설명

PP(생산라인 자재) + FG(완제품 출하) 출고 요청 통합. 시스템이 FIFO 규칙(입고일 오래된 LOT 우선)으로 피킹 위치 자동 추천. [Pick] 버튼 클릭 시 WH-07 PDA 피킹 화면으로 선택 항목 인계.



PP + FG Merged

자재 + 출하 단일 큐.
우선순위 통합



FIFO Auto-Suggest

입고일 오래된 LOT 자동 추천.
BR-WH-04 강제



Due Date Priority

납기 임박 행 노랑·빨강.
자동 정렬



→ WH-07 Trigger

[Pick] 클릭 시 PDA 피킹 진입.
선택 인계

관련 BR ▶

BR-WH-04



WH-07 · CORE

PDA Release Picking

PDA 출고 피킹 — FIFO 강제 + 오폭목 차단

[PDA](#)[Core](#)[L3 DETAIL →](#)

What you'll see: PDA 안내 화면 (시스템이 가장 오래된 LOT 위치로 안내 → 작업자 바코드 확인 → 수량)

FIFO 위반 시도 또는 오폭목 스캔 즉시 차단 (BR-WH-04)

INPUT

Pick list (FIFO sorted)

OUTPUT

Stock OUT + Manifest

BLOCK

Wrong item · FIFO violation

UI

PDA Guided Flow

EN · DESCRIPTION

KO · 설명

PDA-guided FIFO picking for outbound. System suggests oldest stock location first. Operator scans item barcode to confirm correct pick; wrong-item scan immediately blocked. Multi-pick supported with consolidated manifest. Confirms outbound + prints delivery slip.

PDA 안내 기반 FIFO 출고 피킹. 시스템이 입고일 가장 오래된 LOT 위치 자동 제안. 작업자가 품목 바코드 스캔으로 정확성 확인 — 오폐품 즉시 차단. 다중 피킹 지원 + 통합 인보이스. 출고 확정 + 배송 전표 출력.



FIFO Guided

오래된 LOT 위치 자동 안내.
현장 해매기 0



Wrong-Item Block

오폐품 바코드 스캔 즉시 STOP.
잘못된 출고 0



Multi-Pick

한 출고 요청 다중 LOT/위치 일괄.
시간 단축



Manifest Print

배송 전표 자동 인쇄.
PP/FG 인계

관련 BR ▶

BR-WH-04



WH-08

Transaction History

입출고 이력 — 전수 감사 추적

Web PC

1st Dev

L3 DETAIL →



What you'll see: 모든 입출고 · 조정 트랜잭션 통합 그리드 + 다중 필터 (날짜 · 품목 · 작업자 · 사유)

감사 대비 Excel 출력 + 영구 보관 (사유 코드별 색상)

INPUT

All WH events

OUTPUT

Audit Trail + Excel

RETAIN

7 years (감사)

UI

Multi-Filter Grid

EN · DESCRIPTION

Full audit trail of all inbound, outbound, and adjustment transactions: timestamp, operator, quantity, location, reason code, before/after snapshot. Searchable by date range, item, operator, reason. Excel export for audit. Permanent retention 7 years.

KO · 설명

모든 입고 · 출고 · 조정 트랜잭션 감사 기록: 일시 · 작업자 · 수량 · 위치 · 사유 코드 · 전후 스냅샷. 날짜/품목/작업자/사유 검색. 감사 대비 Excel 출력. 7년 영구 보관.



Multi-Filter

날짜 · 품목 · 작업자 · 사유 조합 검색.
시점 추적 가능



Excel Audit Export

감사 대비 전수 출력.
서명 라인 포함



Reason Color Code

사유별 행 색상 차별화.
분석 용이



Immutable

한 번 기록 후 변경 불가.
7년 영구 보관

관련 BR ▶ 전 BR 이력 통합 보관

▼ LOCATION HIERARCHY · 로케이션 체계 (MD-11)

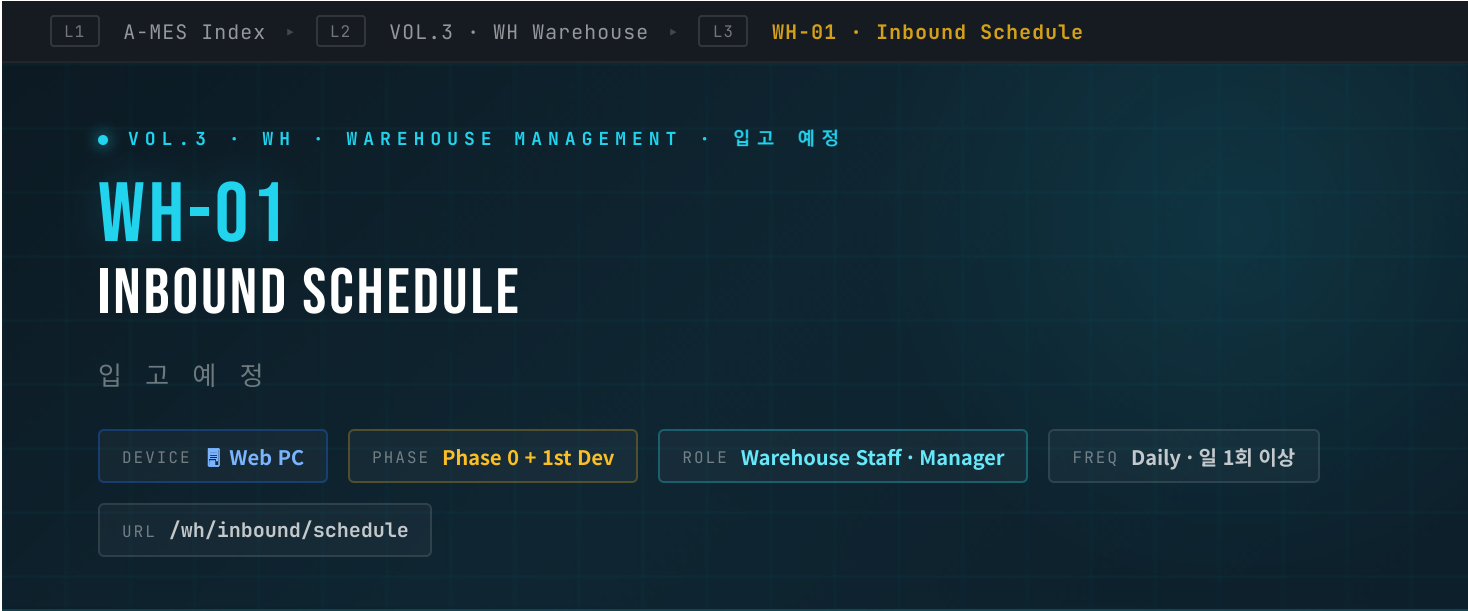
Level	Code Pattern	Example	Description · 설명
Zone · 구역	[A-Z]	A · B · C	창고 대구역 (자재 / 반제품 / 완제품 등).
Row · 행	[1-99]	A-12	구역 내 통로 번호.
Column · 열	[01-99]	A-12-05	행 내 선반 위치.
Level · 층	[01-09]	A-12-05-03	선반 층(1층 = 최하단).
Full ID	{Z}-{R}-{C}-{L}	A-12-05-03	QR 라벨 인쇄 양식. 4단계 계층.

▼ ADJUSTMENT REASON CODES · 재고 조정 사유

Code	Name	KO	When to Use · 사용 시점
DAMAGED	Damaged Stock	파손	이동/보관 중 파손 확인 시. 사진 첨부 권장. → 폐기 사유 코드와 별개.
LOST	Lost Stock	분실	cycle count 결과 시스템 재고보다 실재고가 적을 때.
FOUND	Found Stock	발견	예상 외 위치에서 발견되어 시스템에 반영.
COUNT_DIFF	Cycle Count Diff	실사 차이	정기 실사 후 시스템-실재고 차이 보정.
OTHER	Other (text required)	기타	위 4종에 해당하지 않는 경우. 자유 텍스트 200자 + PIN 필수.

▼ BUSINESS RULES · 업무 규칙 (WH · 4건)

Rule ID	Category	Rule (EN)	규칙 (KO)	Action
BR-WH-01	Inbound Gate	Items without a valid open PO cannot be received. Barcode scan is blocked immediately if no matching PO is found in the system.	PO 없는 품목 입고 불가. 미등록 바코드 스캔 즉시 차단.	Block
BR-WH-02	QC Auto-Trigger	All inbound items automatically trigger QC-01 incoming inspection. Inventory is updated only after QC pass. QC fail → items quarantined.	입고 완료 즉시 QC-01 수입 검사 자동 트리거. 합격 후 재고 반영. 불합격 시 격리.	Auto
BR-WH-03	Min Stock Alert	When on-hand quantity falls below the minimum stock level (defined in MD-09/10), an alert is sent to the warehouse manager and procurement team.	현재고 < 최소재고(MD-09/10 기준) 시 창고담당·구매팀 즉시 알림.	Alert
BR-WH-04	FIFO Picking	All outbound picking must follow FIFO (First In First Out) based on inbound date. System auto-suggests oldest stock location. Wrong-LOT scan blocked.	출고 피킹 입고일 기준 FIFO 자동 적용. 최고령 재고 위치 자동 제안. 다른 LOT 스캔 차단.	Auto



01 Overview

화면 개요

Displays all open purchase orders received from the SCM portal and allows warehouse staff to register expected inbound schedules. ETA D-Day is auto-calculated; overdue deliveries are highlighted in red and trigger alerts to the procurement and warehouse manager. The **[Receive]** action launches the **WH-02 PDA Inbound** flow for the selected PO, while QC-01 incoming inspection is auto-queued upon confirmation.

SCM 포털에서 수신한 모든 미입고 발주(PO)를 표시하고, 창고 담당자가 입고 예정 일정을 등록할 수 있는 화면입니다. ETA D-Day는 자동 계산되며, 지연 발주는 빨간색으로 강조되고 구매팀·창고 관리자에게 알림이 발송됩니다. **[Receive]** 클릭 시 선택한 PO에 대해 **WH-02 PDA 입고** 흐름이 시작되며, 확정 즉시 QC-01 수입검사가 자동 대기열에 등록됩니다.

SCREEN ID	WH-01
MODULE	VOL.3 · Warehouse 창고 관리
LAYOUT	Toolbar + KPI strip + Data table 툴바·KPI·표
REFRESH	Auto every 30 min + manual 30분 자동 + 수동
PRIMARY KPI	Pending PO, Arriving Today, Overdue
TABLES	wh_inbound , po_header , md_item
RBAC	WH.View · WH.Receive

02 Screen Mockup

화면 목업

A-MES · Warehouse · Inbound Schedule

Warehouse > Inbound > Schedule

All Status ▾

All Vendors ▾

Search PO No. / Item...

연도 - 월 - 일

~

연도 - 월 - 일

Search

+ Register Schedule

Excel

12

Pending PO
입고 예정 건

3

Arriving Today
금일 입고

1

Overdue
지연

5

This Week
이번 주

3

Next Week
다음 주

PO No. 발주번호	Vendor 공급사	Item Code 품목코드	Item Name 품목명	Order Qty 발주 수량	Unit 단위	ETA 입고 예정 일	D-Day	Status 상태	Action
PO-2025-0628-003	삼성화학	RM-PP-NAT-20	PP Pellet Natural 20MFI PP 펠렛 나투럴	500	kg	2025-06-28	D+3	Overdue · 지연	Receive
PO-2025-0701-001	현대원단	FAB-GY-C03	Fabric Roll Grey GY-C03 원단 롤 회색	20	roll	2025-07-01	Today	Arriving · 도착 예정	Receive
PO-2025-0705-002	KCC도료	PNT-BK-NH578	Black Paint NH578 블랙 도료 NH578	50	L	2025-07-05	D-4	Scheduled · 예정	Detail · 상세
PO-2025-0708-004	현대원단	FAB-BK-C01	Fabric Roll Black BK-C01 원단 롤 블랙	10	roll	2025-07-08	D-7	Scheduled · 예정	Detail · 상세
PO-2025-0710-005	삼성화학	RM-PP-BLK-20	PP Pellet Black 20MFI PP 펠렛 블랙	300	kg	2025-07-10	D-9	Scheduled · 예정	Detail · 상세

Row color legend · 행 색상 — Red row: Overdue PO · 지연 / Green row: Arriving today · 금일 도착 / White: Scheduled · 예정. D-Day 색상: D+N (red) · 지연, Today (green) · 금일, D-N (default) · 예정.

03

UI Component Specification

UI 컴포넌트 명세

3.1 Toolbar Components · 툴바 컴포넌트

ID	COMPONENT 컴포넌트	TYPE 타입	REQ 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT 기본값
CMP-01	Status Filter 상태 필터	SELECT	N	Options: All / Pending / Arriving Today / Overdue / Completed . Changes trigger table reload.	All Status
CMP-02	Vendor Filter 공급사 필터	SELECT	N	Loaded from md_vendor where is_active=1 . Multi-select disabled in v1.	All Vendors
CMP-03	Search Box 검색창	TEXT	N	Min 2 chars. Searches po_no and item_code/name with LIKE %query% . Debounced 300ms.	(empty)
CMP-04	Date Range 기간 선택	DATE × 2	N	From ≤ To. Max range 90 days. Filter on eta_date .	Today ~ +14d
CMP-05	Search Button 검색 버튼	BUTTON	N	Triggers GET /api/wh/inbound/schedule with all filter params. Enter key in any filter also triggers.	—
CMP-06	Register Schedule 예정 등록	BUTTON	N	Opens modal for manual schedule entry (vendor-driven PO without SCM sync). Requires WH.Manage role.	—
CMP-07	Excel Export 엑셀 출력	BUTTON	N	Exports current filtered result as .xlsx . Max 10,000 rows. Filename: WH01_Inbound_YYYYMMDD.xlsx .	—

3.2 KPI Strip · KPI 영역

ID	METRIC 지표	TYPE	CALCULATION 산출 로직	REFRESH
KPI-01	Pending PO 입고 예정 건	COUNT	Count of po_header where status IN ('Scheduled', 'Arriving', 'Overdue') .	30 min
KPI-02	Arriving Today 금일 입고	COUNT	Count where eta_date = CURRENT_DATE . Color: #22c55e .	30 min
KPI-03	Overdue 지연	COUNT	Count where eta_date < CURRENT_DATE AND status ≠ 'Received' . Color: #ef4444 .	30 min
KPI-04	This Week 이번 주	COUNT	Count where eta_date within current ISO week. Excludes Today.	30 min
KPI-05	Next Week 다음 주	COUNT	Count where eta_date within next ISO week.	30 min

3.3 Data Table Columns · 데이터 표 컬럼

ID	COLUMN 컬럼	SOURCE 소스	FORMAT / RULE 표시·규칙	SORT
COL-01	PO No. 발주번호	po_header.po_no	Format: PO-YYYY-MMDD-NNN . Overdue: red bold. Click → PO detail popup.	DESC
COL-02	Vendor 공급사	md_vendor.name	Display vendor short name. Tooltip = full legal name.	ASC
COL-03	Item Code 품목코드	md_item.item_code	Monospace font. Click → item master popup.	ASC
COL-04	Item Name 품목명	md_item.item_name_en/ko	EN main + KO subtext (smaller, muted).	—
COL-05	Order Qty 발주 수량	po_header.order_qty	Right-align. Thousands separator. Decimal: 0-3 digits per unit.	—
COL-06	Unit 단위	md_item.unit	kg / roll / L / EA.	—
COL-07	ETA 입고 예정일	po_header.eta_date	Format: YYYY-MM-DD . Editable inline by Manager role.	ASC ★
COL-08	D-Day 잔여일	Calculated	eta_date - CURRENT_DATE . Show D+N (red) / Today (green) / D-N .	—
COL-09	Status 상태	po_header.status	Badge: Scheduled (blue) / Arriving (yellow) / Overdue (red) / Received (green).	—
COL-10	Action 액션	Button	[Receive] if status in (Arriving / Overdue / Scheduled within 1 day) · else [Detail].	—

04 Event & Action Flow

이벤트 · 액션 흐름

1

SCHEDULED scm.sync.cron @ */30min

System fetches new/updated POs from SCM portal every 30 minutes via SAP B1 SLD adapter.

시스템이 SCM 포탈에서 신규/갱신 PO를 30분마다 SAP B1 SLD 어댑터로 수신합니다.

EFFECT UPSERT into po_header · UPDATE last_synced_at · trigger overdue re-evaluation.

2

USER `page.load` · WH-01 진입

Warehouse staff opens the screen. System auto-applies default filter (Today ~ +14d, All Status).

참고 담당자가 화면에 진입하면 기본 필터(오늘~+14일, 전체 상태)가 자동 적용됩니다.

API `GET /api/wh/inbound/schedule?from=...&to=...&status=all` → KPI strip + table populated.

3

USER `click` · [ Search]

User adjusts filter (status / vendor / date range / search box) and clicks Search. Same effect when pressing Enter in any filter field.

필터(상태/공급사/기간/검색) 변경 후 [검색] 클릭 또는 Enter 입력 시 동일 동작.

VALIDATE `date_from ≤ date_to`, range ≤ 90 days, search ≥ 2 chars → re-query & redraw table.

4

USER `click` · [ Receive]

User clicks Receive on a row. System checks RBAC WH.Receive and PO status validity, then launches WH-02 PDA flow with PO context.

[Receive] 클릭 시 RBAC 권한·PO 상태 검증 후 WH-02 PDA 입고 흐름으로 PO 정보와 함께 전환됩니다.

REDIRECT `POST /api/wh/inbound/start` → returns `session_id` → redirect `/wh/pda/inbound?po=...&sid=...`.

5

SYSTEM `job.overdue.check @ 09:00 daily`

Daily 09:00 KST job re-evaluates ETA. POs past ETA without receipt become Overdue → alert sent to procurement & WH manager via in-app + email.

매일 09:00 KST 배치가 ETA를 재평가하여 미입고 PO를 지연 상태로 전환 + 구매팀·창고관리자에게 알림 발송.

EFFECT `UPDATE po_header SET status='Overdue'` · INSERT into `sys_alert` · email via SYS-04.

6

USER `click` · [+ Register Schedule]

Manager opens modal to manually register an inbound schedule for non-SCM POs (e.g. local consumables, sample shipments).

SCM 외 발주(소모성·샘플 등)에 대해 매니저가 수동으로 입고 예정을 등록할 수 있는 모달을 엽니다.

REQUIRE Role = `WH.Manage` · fields: Vendor, Item, Qty, ETA, Note → `POST /api/wh/inbound/manual`.

✓

SYSTEM `return` · from WH-02 success

After WH-02 PDA inbound completion, user returns to WH-01. Received PO row disappears (or shows Received badge depending on filter).

WH-02 PDA 입고 완료 후 WH-01로 복귀하면 입고 완료된 PO 행이 사라지거나 (필터에 따라) [Received] 상태 배지로 표시됩니다.

DOWNSTREAM QC-01 incoming inspection auto-queued · min-stock alert re-evaluated · WH-08 transaction log written.

05

Exception & Error Handling

예외 · 에러 처리

E-WH01-001

SCM Sync Failure

SCM 동기화 실패

CAUSE	SAP B1 SLD endpoint unreachable / timeout > 30s. SAP B1 SLD 연결 실패·30초 초과.
DETECT	Last <code>last_synced_at</code> > 60 min ago.
ACTION	Show yellow banner at top: "Last sync: HH:MM. Connection issue." Retry button available. Cached data still displayed.
NOTIFY	SYS admin via in-app alert + email after 3 consecutive failures.

⚠ SCM SYNC DELAYED — DATA MAY BE STALE

E-WH01-002

Receive Without Permission

권한 없는 입고 시도

CAUSE	User clicks [Receive] without <code>WH.Receive</code> role. 권한 없는 사용자가 입고 시도.
DETECT	Backend RBAC check on <code>POST /api/wh/inbound/start</code> .
ACTION	Block API call. Return <code>403 Forbidden</code> . Modal: "You don't have permission to receive. Contact your manager."
LOG	Record attempt in <code>sys_audit</code> with <code>user_id</code> , <code>timestamp</code> , <code>attempted_po</code> .

🚫 ACCESS DENIED — ROLE WH.RECEIVE REQUIRED

E-WH01-003

PO Already Received

이미 입고된 PO

CAUSE	Another user already completed WH-02 for this PO (race condition). 동시 작업으로 이미 다른 사용자가 입고 완료.
DETECT	On [Receive] click: <code>SELECT status FROM po_header</code> → <code>status = 'Received'</code> .
ACTION	Refuse navigation to WH-02. Show toast: "This PO has been received by [user] at [time]." Refresh table.
RECOVERY	User views WH-08 Transaction History for confirmation.

⚠ ALREADY RECEIVED — SEE WH-08 HISTORY

E-WH01-004

Invalid Date Range

잘못된 기간 입력

CAUSE	<code>date_from > date_to</code> or range > 90 days. 시작일 > 종료일 또는 기간 90일 초과.
DETECT	Client-side validation before API call.
ACTION	Highlight date inputs with red border. Inline error: "Range must be valid and ≤ 90 days." Search button disabled.
RECOVERY	User corrects dates → validation passes → button re-enabled.

⚠ INVALID RANGE — MAX 90 DAYS

E-WH01-005

Excel Export Overflow

엑셀 출력 초과

CAUSE	Filtered result > 10,000 rows. 필터 결과가 1만 행 초과.
DETECT	Server-side count check before file generation.
ACTION	Reject export. Modal: "Result exceeds 10,000 rows. Narrow your filter or use Reports module."
RECOVERY	User reduces date range or applies vendor filter, then retries.

⚠ TOO MANY ROWS — NARROW FILTER

E-WH01-006

Empty Result Set

조회 결과 없음

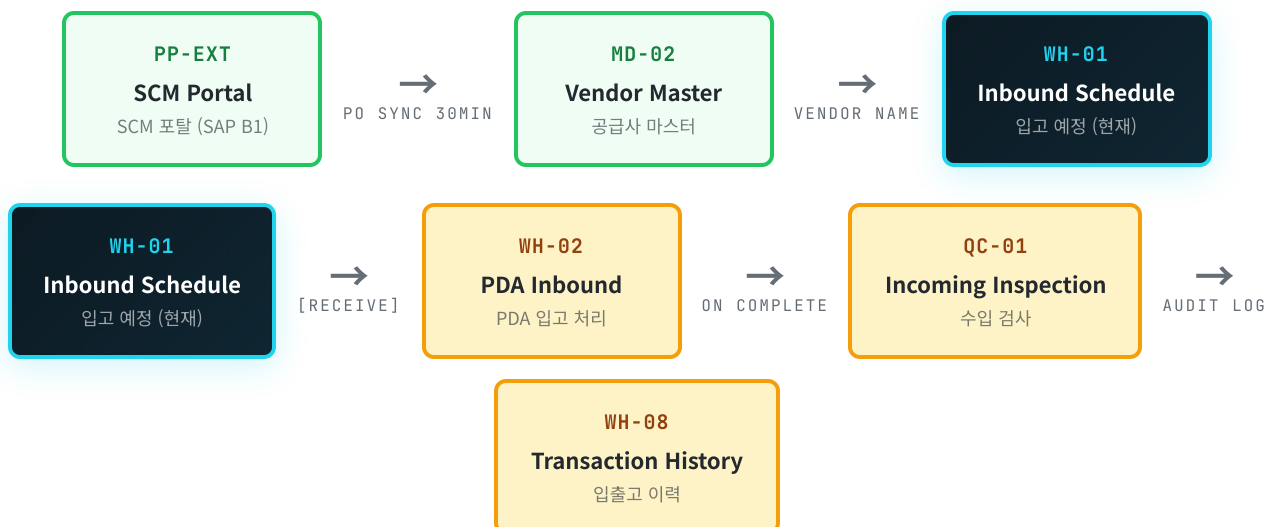
CAUSE	No PO matches current filter. 필터 조건에 맞는 발주 없음.
DETECT	API returns empty array.
ACTION	Show empty state illustration + message: "No inbound schedules match your filter." KPI strip shows zeros but remains visible.
RECOVERY	Reset filter button shown.

ⓘ NO RESULTS — TRY CLEARING FILTER

06

Linked Screens

선행 · 후행 연계 화면

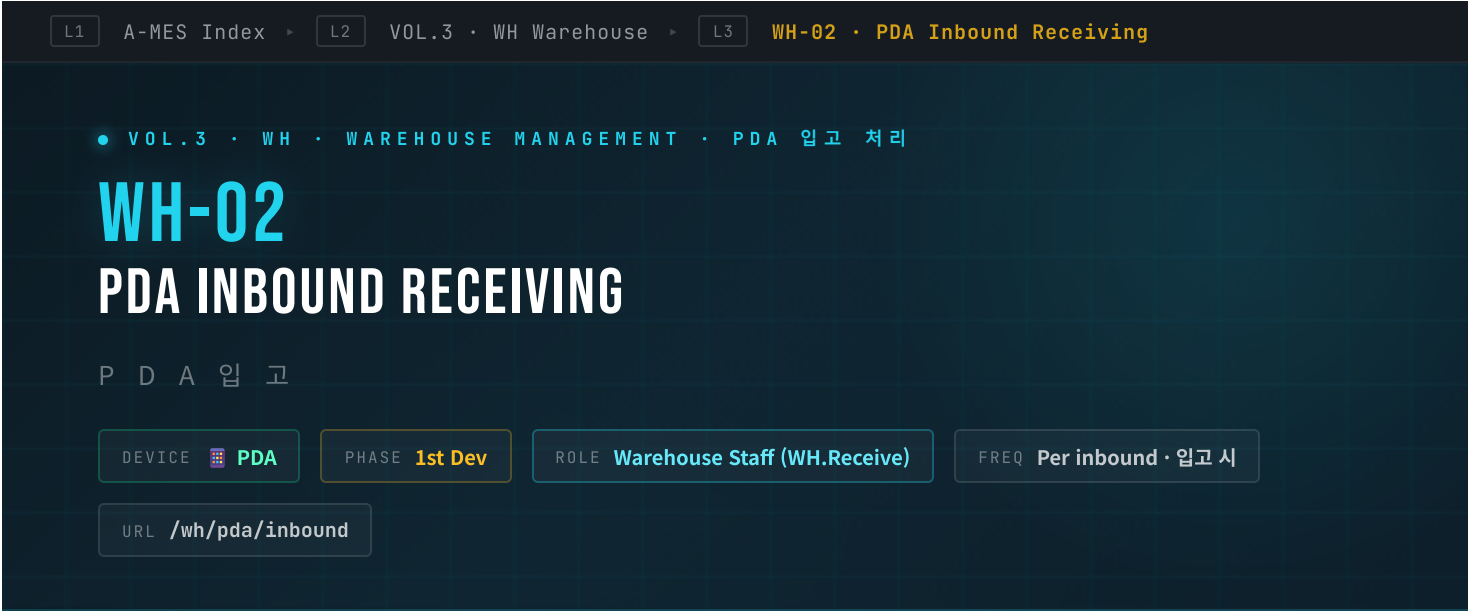


RULE ID	DESCRIPTION	ACTION
BR-WH-01	Items without a valid open PO cannot be received. Barcode scan is blocked immediately if no matching PO is found in the system. 유효한 미입고 PO가 없는 품목은 입고할 수 없으며, 일치하는 PO가 없는 바코드 스캔은 즉시 차단됩니다.	Block
BR-WH-02	All inbound items automatically trigger QC-01 incoming inspection. Inventory is updated only after QC pass. 모든 입고 품목은 QC-01 수입검사가 자동 트리거되며, 검사 합격 후에만 재고에 반영됩니다.	Auto
BR-WH-03	When on-hand quantity falls below minimum stock level (MD-09/10), an alert is sent to the warehouse manager and procurement team. 현재고가 최소재고(MD-09/10) 미만이면 창고 관리자·구매팀에 즉시 알림 발송.	Alert

✦

Note · 비고

— BR-WH-04 (FIFO) does not apply to inbound flow; it governs outbound picking (WH-07). For full WH business rules, refer to **VOL.3 WH module — Business Rules section.**



01

Overview

화면 개요

PDA-driven receiving workflow consisting of three sequential steps: **(1) Scan item barcode** against the open PO, **(2) Confirm quantity** via numeric keypad, and **(3) Scan shelf QR code** to assign storage location. The system validates each step in real time — barcode mismatches against the PO item are *immediately blocked*, and the worker cannot proceed until correction. Upon confirmation, inventory is incremented, a location label is printed, and the QC-01 incoming inspection queue is auto-triggered.

PDA 기반 3단계 입고 처리: **(1) 품목 바코드 스캔**으로 PO와 대조, **(2) 숫자패드로 수량 확인**, **(3) 선반 QR 스캔**으로 로케이션 지정. 각 단계에서 실시간 검증되며, PO 품목과 불일치 시 즉시 차단되어 수정 전까지 진행 불가. 확정 시 재고 증가·위치 라벨 출력·QC-01 수입검사 큐 자동 등록이 이루어집니다.

SCREEN ID	WH-02
ENTRY	From WH-01 [Receive] button WH-01 입고 클릭
LAYOUT	3-step wizard · full-screen 3단계 위저드
HARDWARE	Zebra TC22 · 1D/2D scanner
NETWORK	WiFi · offline buffer 4hr 오프라인 4시간 버퍼
TABLES	wh_inbound , wh_inventory , po_header
RBAC	WH.Receive

02

Screen Mockup

화면 목업 — PDA 3단계

① SCAN ITEM · 품목 스캔

WH-02 · Inbound 09:12

Inbound Receiving

입고 처리 · PO-2025-0701-001



Scan Item Barcode
품목 바코드 스캔

PO NO.

P0-2025-0701-001

EXPECTED · 예정 품목

FAB-GY-C03

VENDOR

현대원단

② CONFIRM QTY · 수량 입력

WH-02 · Inbound 09:13

Confirm Quantity

수량 확인 · FAB-GY-C03 ✓

ITEM VERIFIED ✓

FAB-GY-C03 · Grey Fabric

EXPECTED · 예정 수량

20 rolls

20

1

2

3

4

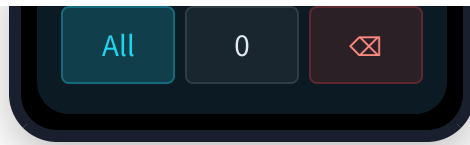
5

6

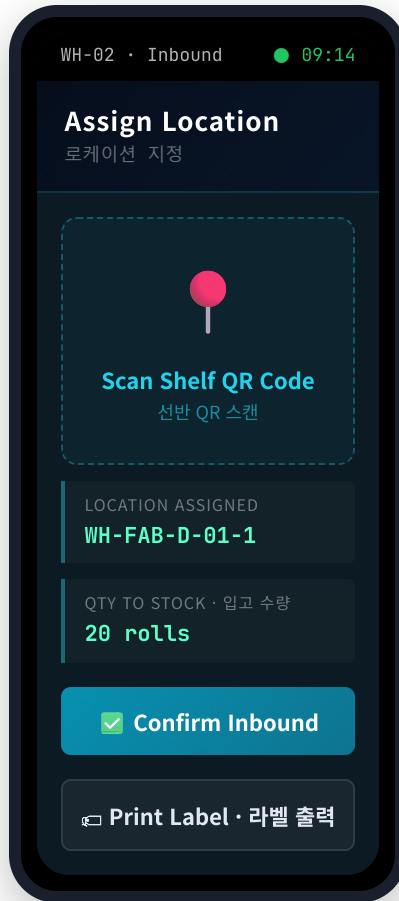
7

8

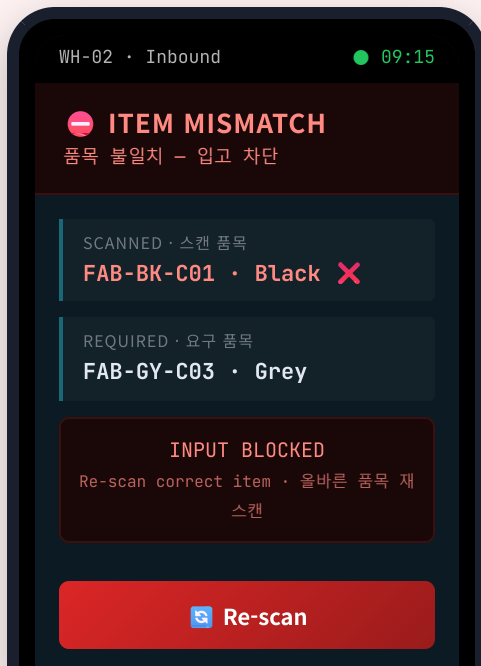
9



③ ASSIGN LOCATION · 위치 지정



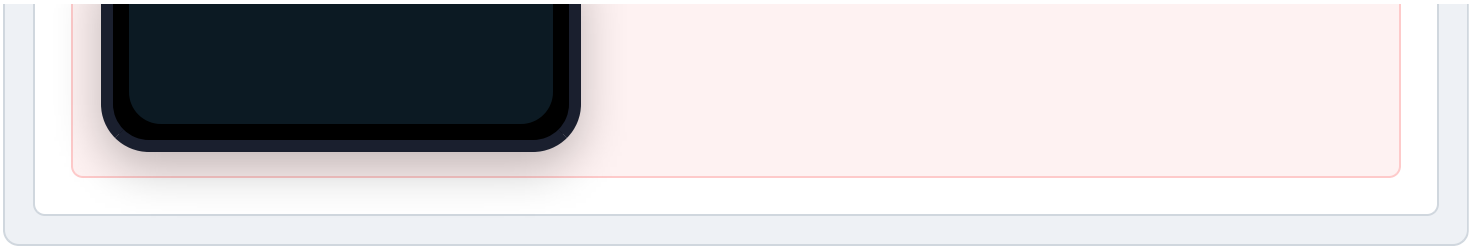
🚫 Error Case · 에러 케이스 — Item Mismatch · 품목 불일치



Trigger · 발생 시점 — Step ① 품목 바코드가 PO 품목과 불일치 시.

Action · 동작 — 입력 즉시 차단 · 빨간 화면 + 진동 + 비프음 알림 · 재스캔 버튼만 활성화.

Recovery · 복구 — 올바른 품목으로 재스캔 시 정상 흐름 복귀. 3회 연속 실패 시 관리자 알림 발송.



03

UI Component Specification

UI 컴포넌트 명세

3.1 Step ① — Item Scan · 품목 스캔

ID	COMPONENT 컴포넌트	TYPE	REQ	VALIDATION / BEHAVIOR 검증 · 동작	ON SUCCESS
CMP-01	Barcode Scanner 바코드 스캐너	SCAN	Y	Reads 1D/2D barcode (Code128, QR). Match against <code>po_header.item_code</code> . Mismatch → block.	→ Step ②
CMP-02	PO Display PO 표시	LABEL	—	Read-only. Shows <code>po_no</code> , expected item code, vendor name from launch context.	—
CMP-03	Manual Entry 수동 입력	BUTTON	N	Fallback if barcode damaged. Opens keyboard input. Same validation as scanner. Logs <code>manual_entry=true</code> in audit.	→ Step ②

3.2 Step ② — Quantity Confirmation · 수량 확인

ID	COMPONENT 컴포넌트	TYPE	REQ	VALIDATION / BEHAVIOR	ON SUCCESS
CMP-04	Numeric Keypad 숫자패드	NUMPAD	Y	0-9, decimal point (per unit precision), backspace. Max 6 digits + 3 decimals.	→ Step ③
CMP-05	[All] Button 전량버튼	BUTTON	N	Auto-fills with full expected quantity from PO. Shortcut for common case.	→ Step ③
CMP-06	Qty Display 수량표시	DISPLAY	-	Large monospace digits. Color: cyan if within tolerance $\pm 2\%$ of expected, yellow $\pm 10\%$, red $>10\%$.	-
CMP-07	Partial Inbound Toggle 부분 입고 토글	SWITCH	N	If qty < expected, this auto-enables. PO status becomes Partial , remaining qty stays open.	-

3.3 Step ③ — Location Assignment · 위치 지정

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	ON SUCCESS
CMP-08	Shelf QR Scanner 선반 QR 스캐너	SCAN	Y	Reads location QR. Format: [Warehouse]-[Zone]-[Row]-[Col]-[Level] . Check capacity from md_location .	→ Confirm
CMP-09	Suggested Location 권장 위치	LABEL	-	System suggests location based on item type + nearest available cell with capacity. Worker can override.	-
CMP-10	[Confirm Inbound] 입고 확정	BUTTON	Y	Commits transaction. Increments wh_inventory.on_hand_qty . Inserts wh_inbound . Triggers QC-01.	→ Print
CMP-11	[Print Label] 라벨 출력	BUTTON	N	Prints location label via Zebra ZD230 over Bluetooth. Includes item, qty, location, inbound date.	-

1

USER launch · from WH-01 [Receive]

User clicks [Receive] on a PO row in WH-01. PDA launches WH-02 with PO context. Step ① is shown.

WH-01에서 [Receive] 클릭 시 WH-02가 PO 정보와 함께 실행되며 Step ① 화면 표시.

CONTEXT `session_id`, `po_no`, `expected_item` loaded into PDA local state.

2

PDA SCAN trigger · item barcode

Worker pulls trigger to scan item barcode. System matches against PO expected item.

작업자가 트리거를 당겨 품목 바코드 스캔. 시스템이 PO 예정 품목과 대조.

VALIDATE `scanned_code = po.item_code` → success (green) · mismatch → block (E-WH02-001).

3

USER tap · numpad / [All]

Worker enters received quantity via numpad or taps [All] to auto-fill expected quantity. Display shows qty with tolerance color coding.

숫자패드 입력 또는 [All] 버튼으로 수량 입력. 디스플레이는 허용오차 색상 표시.

DETECT `qty < expected` → auto-enable Partial Inbound mode. `qty > expected` → warn (E-WH02-003).

4

PDA SCAN trigger · shelf QR

Worker scans the shelf QR code at the target storage location. System validates capacity vs incoming qty.

대상 선반의 QR 코드를 스캔. 시스템이 용량 대비 입고량 검증.

VALIDATE `location.capacity - location.on_hand ≥ qty` → OK · else block (E-WH02-004).

5

USER tap · [Confirm Inbound]

Worker confirms. Backend commits transaction atomically: insert inbound, update inventory, update PO status, queue QC-01.

작업자 확정. 백엔드가 트랜잭션을 원자적으로 커밋: 입고 INSERT, 재고 UPDATE, PO 상태 변경, QC-01 큐 등록.

API `POST /api/wh/inbound/confirm` · `BEGIN TRAN ... COMMIT` · returns `inbound_id`.

6

SYSTEM print · location label

Optionally prints a location label via Bluetooth-paired Zebra ZD230 printer. Worker affixes label to the lot.

블루투스 페어링된 Zebra ZD230 프린터로 위치 라벨 출력(선택). 작업자가 로트에 부착.

OUTPUT ZPL command via Bluetooth · label contains item, qty, location, date, `inbound_id` barcode.

✓

SYSTEM complete · return to WH-01

PDA returns to WH-01. Received PO is no longer in pending list. QC-01 inspector receives notification.

PDA가 WH-01로 복귀. 처리된 PO는 미입고 목록에서 제외. QC-01 검사자에게 알림.

DOWNSTREAM QC-01 queue · WH-03 inventory updated · WH-08 transaction log · min-stock check.

05 Exception & Error Handling

예외 · 에러 처리

E-WH02-001 Item Barcode Mismatch 품목 바코드 불일치

CAUSE	Scanned item code $\neq$ PO expected item. 스캔 품목이 PO 예정 품목과 다름.
DETECT	Step ① real-time comparison.
ACTION	Red full-screen warning + vibration + beep. Re-scan button only. 3 consecutive failures $\rightarrow$ manager alert.
RECOVERY	Re-scan correct item to resume.

⊖ ITEM MISMATCH — INPUT BLOCKED

E-WH02-002 PO Not Found / Expired PO 없음 · 만료

CAUSE	PO context invalid: cancelled, already received, or session timeout. PO 취소 · 이미 입고 · 세션 만료.
DETECT	Server check on Step ① scan.
ACTION	Block scan. Modal: "PO no longer valid. Return to WH-01." Auto-redirect after 5 sec.
RECOVERY	Re-select PO from WH-01.

⊖ PO INVALID — RETURN TO LIST

E-WH02-003 Qty Exceeds PO 발주량 초과

CAUSE	Entered qty > expected by >5%. 입력 수량이 발주 수량 5% 초과.
DETECT	Step ② numeric validation.
ACTION	Yellow display. Confirmation popup: "Receive QTY (over by N%)? Comment required." Free-text reason input.
RECOVERY	User confirms with reason, or corrects qty.

⚠ OVER-RECEIPT — REASON REQUIRED

E-WH02-004 Location Capacity Exceeded 로케이션 용량 초과

CAUSE	Target location's free capacity < incoming qty. 대상 위치 잔여 용량 부족.
DETECT	Step ③ on QR scan.

ACTION Block confirmation. Show free capacity. Suggest alternative location.

RECOVERY Scan different location, or split into multiple locations.

— LOCATION FULL — SCAN ANOTHER

E-WH02-005

Network Disconnected

네트워크 단절

CAUSE WiFi signal lost or backend unreachable.
WiFi 단절 또는 백엔드 응답 없음.

DETECT PDA heartbeat check every 10s.

ACTION Switch to offline mode. Queue transactions locally (up to 4hr). Yellow banner. Auto-sync on reconnect.

RECOVERY Reconnect to WiFi → queued transactions sync in FIFO order.

△ OFFLINE — QUEUE MODE ACTIVE

E-WH02-006

Printer Not Ready

프린터 미응답

CAUSE Bluetooth printer disconnected or out of paper.
블루투스 단절·용지 없음.

DETECT Print request timeout 8s.

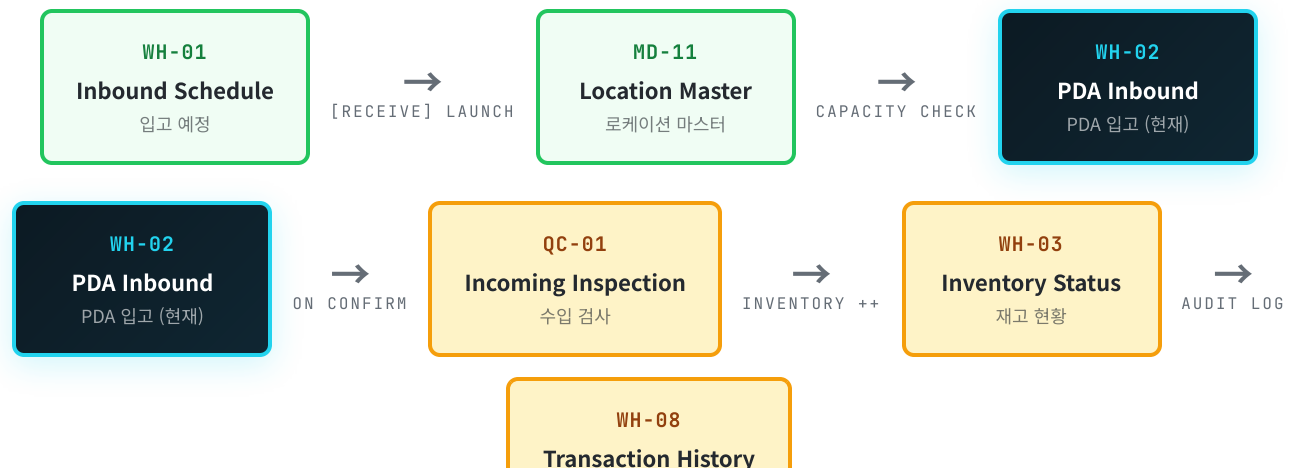
ACTION Inbound transaction NOT rolled back. Show: "Label print failed. Retry?" Inventory commit already complete.

RECOVERY Re-print from WH-08 transaction history.

△ PRINT FAILED — RETRY FROM WH-08

06 Linked Screens

선행 · 후행 연계 화면

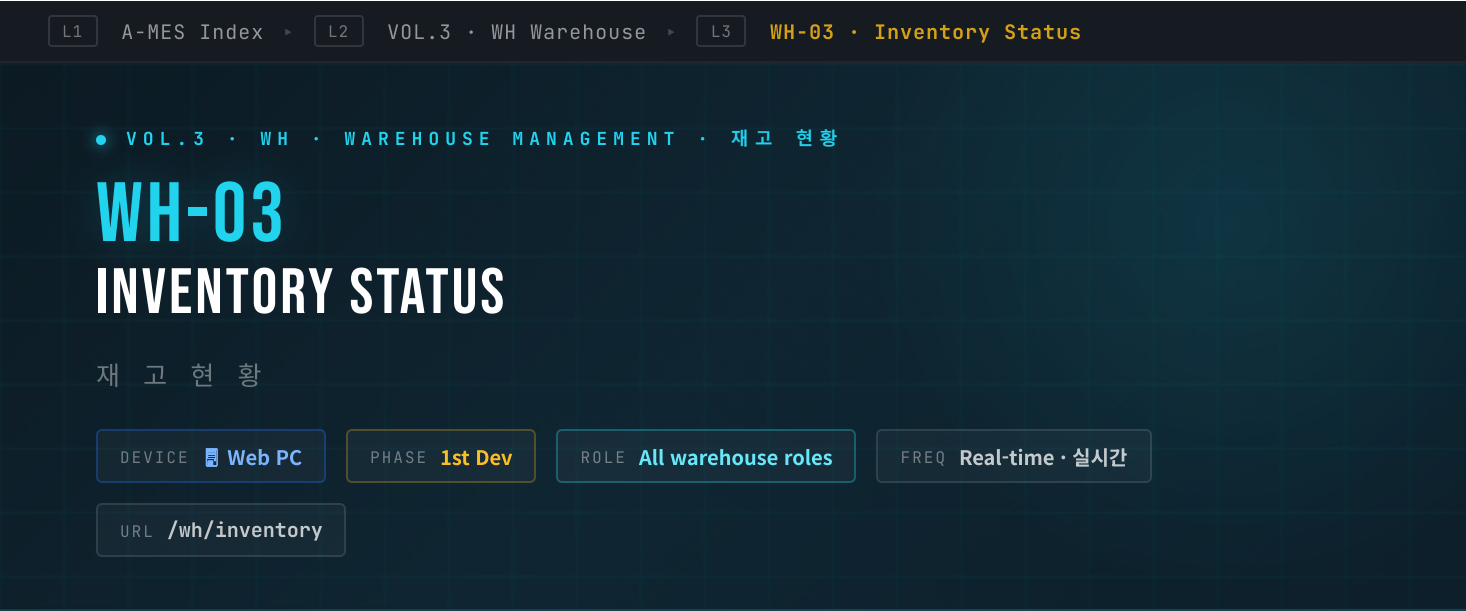


07 Applicable Business Rules

적용 업무 규칙

RULE ID	DESCRIPTION	ACTION
BR-WH-01	Items without a valid open PO cannot be received. Barcode scan blocked immediately if no matching PO is found. 유효한 미입고 PO 없는 품목 입고 불가. 미일치 바코드 즉시 차단.	Block
BR-WH-02	All inbound items automatically trigger QC-01 incoming inspection. Inventory updated only after QC pass. 모든 입고 품목 QC-01 자동 트리거. 검사 합격 후 재고 반영.	Auto
BR-WH-03	When on-hand qty falls below minimum stock level (MD-09/10), an alert is sent to manager and procurement. 최소재고 미달 시 창고관리자·구매팀 알림 발송.	Alert

 **Note · 비교** — WH-02 is the only entry point for inbound transactions. Manual inventory entry must use WH-05 (Adjustment), which requires supervisor approval. WH-02는 입고 거래의 유일한 진입점이며, 수동 입력은 WH-05 (조정)로 처리하고 관리자 승인이 필요합니다.



01

Overview

화면 개요

Real-time inventory dashboard showing stock by item, material type (Raw Material / Fabric / Paint), and location. Items below the minimum stock level (from MD-09 · Min/Max Stock Master) are highlighted in red and surface in the KPI strip. The screen supports Excel export of the filtered view and shows daily snapshot delta vs. the previous day's closing inventory, so staff can quickly spot abnormal consumption or replenishment needs.

품목·유형(원자재/원단/도료)·로케이션별 실시간 재고 현황 대시보드입니다. 최소재고(MD-09 · 최소·최대재고 마스터) 미달 항목은 빨간색으로 강조되고 KPI에 별도 집계됩니다. 필터링된 결과의 엑셀 출력을 지원하며, 전일 마감 재고 대비 증감을 표시하여 비정상 소비·보충 필요 항목을 빠르게 식별할 수 있습니다.

SCREEN ID	WH-03
MODULE	VOL.3 · Warehouse 창고 관리
LAYOUT	Filter bar + KPI + Data table 필터·KPI·표
REFRESH	Real-time push (WebSocket) 실시간 푸시
SNAPSHOT	Daily 00:00 KST 매일 자정 스냅샷
TABLES	wh_inventory , md_item , md_location , md_minmax
RBAC	WH.View (read-only for most users)

02

Screen Mockup

화면 목업

A-MES · Warehouse · Inventory Status

Warehouse > Inventory

All Types



All Warehouses



All Status



Item Code / Name...

Search

Export Excel

Refresh

124

Total Items
전체 품목 수

3

△ Below Min
최소재고 미달

23

Low Stock
재고 부족 주의

98

Normal
정상

₩ 42.7M

Total Value
총 재고 금액

Item Code 품목코드	Item Name 품목명	Type 유형	On Hand 현재고	Unit	Min Stock 최소재고	Δ vs Yesterday 전일 대비	Location 위치	Status
RM-PP-NAT-20	PP Pellet Natural 20MFI PP 펠렛 나투럴	Raw Material	1,250 kg	kg	300 kg	▲ 500 kg	WH-RM-A-01-2	Normal
FAB-GY-C03	Fabric Roll Grey GY-C03 원단 롤 회색	Fabric	42 roll	roll	15 roll	▲ 20 roll	WH-FAB-D-01-1	Normal
PNT-BK-NH578	Black Paint NH578 블랙 도료 NH578	Paint	284 L	L	100 L	▼ 16 L	WH-PNT-E-01-1	Normal
FAB-BK-C01	Fabric Roll Black BK-C01 원단 롤 블랙	Fabric	18 roll	roll	15 roll	▼ 4 roll	WH-FAB-D-02-1	Low Stock
RM-MB-GY-03	Color Masterbatch Grey 착색 마스터배치 회색	Raw Material	0.5 kg	kg	2 kg	▼ 0.3 kg	WH-RM-A-02-1	△ Below Min

Row legend · 행 색상 — Green: Normal / Yellow: Low Stock (within 30% of min) / Red: Below minimum stock level. Δ column shows change since previous-day 00:00 snapshot.

03 UI Component Specification

UI 컴포넌트 명세

3.1 Filter Toolbar · 필터 톨바

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	DEFAULT
CMP-01	Type Filter 유형 필터	SELECT	N	Options: All / Raw Material / Fabric / Paint / WIP . Multi-select disabled v1.	All Types
CMP-02	Warehouse Filter 창고 필터	SELECT	N	Loaded from md_warehouse . WH-RM / WH-FAB / WH-PNT / WH-FG.	All Warehouses
CMP-03	Status Filter 상태 필터	SELECT	N	Options: All / Below Min / Low Stock / Normal / Over Max . Below Min auto-applied on KPI click.	All Status
CMP-04	Search Box 검색창	TEXT	N	Min 2 chars. Search by item_code or item_name . Debounce 300ms.	(empty)
CMP-05	Refresh 새로고침	BUTTON	N	Force-fetch latest data ignoring WebSocket buffer.	—
CMP-06	Excel Export 엑셀 출력	BUTTON	N	Exports current filter result. Max 10,000 rows. Filename: WH03_Inventory_YYYYMMDD_HHmm.xlsx .	—

3.2 KPI Strip · KPI

ID	METRIC	TYPE	CALCULATION	DRILL-DOWN
KPI-01	Total Items 전체 품목 수	COUNT	COUNT(DISTINCT item_code) FROM wh_inventory WHERE on_hand_qty > 0 .	—
KPI-02	Below Min 최소재고 미달	COUNT	Count where on_hand_qty < md_minmax.min_qty . Color: #dc2626 . Click → filter status=BelowMin.	Filter apply
KPI-03	Low Stock 재고 부족	COUNT	Count where on_hand_qty within 1.0× ~ 1.3× min_qty. Color: #f59e0b .	Filter apply
KPI-04	Normal 정상	COUNT	Count where on_hand_qty ≥ 1.3 × min_qty . Color: #22c55e .	Filter apply
KPI-05	Total Value 총 재고 금액	SUM	SUM(on_hand_qty × unit_cost) from md_item.unit_cost . Display in W.	—

3.3 Data Table Columns · 데이터 표 컬럼

ID	COLUMN	SOURCE	FORMAT / RULE	SORT
COL-01	Item Code 품목코드	md_item.item_code	Monospace. Below-min items: bold red. Click → item master popup.	ASC
COL-02	Item Name 품목명	md_item.name_en/ko	EN main + KO subtext.	—
COL-03	Type 유형	md_item.type	Badge: RM (gray) / Fabric (blue) / Paint (orange) / WIP (purple).	—
COL-04	On Hand 현재고	wh_inventory.on_hand_qty	Right-align. Below min: red. Low: yellow. Normal: green. Decimal per unit.	DESC
COL-05	Unit 단위	md_item.unit	kg / roll / L / EA.	—
COL-06	Min Stock 최소재고	md_minmax.min_qty	Right-align. Click → edit modal (Manager only).	—
COL-07	Δ vs Yesterday 전일 대비	Calculated	on_hand - snapshot_prev_day . ▲ green if positive, ▼ red if negative.	—
COL-08	Location 위치	wh_inventory.location_id	If multiple locations, show count: WH-FAB-D-01 (+2) . Click → WH-04 map.	—
COL-09	Status 상태	Calculated	Auto-derived from on_hand vs min/max thresholds.	—

04 Event & Action Flow

이벤트 · 액션 흐름

1

USER

page.load · WH-03 진입

User opens WH-03. Default filter (All types, all warehouses) applied. WebSocket subscribes to inventory updates.

WH-03 진입 시 기본 필터 적용 + WebSocket 재고 업데이트 구독.

API

GET /api/wh/inventory + WS /ws/inventory.subscribe

2

SYSTEM

push · inventory.updated

When WH-02 inbound or WH-07 release commits, server pushes inventory delta to all subscribed clients. Row updates in place with brief highlight.

WH-02 입고·WH-07 출고 커밋 시 서버가 재고 변경분을 구독 클라이언트에 푸시. 해당 행이 잠시 하이라이트되며 갱신.

PAYLOAD

{item_code, location_id, on_hand_qty, delta, timestamp}

USER

click · KPI [Below Min]

3

User clicks a KPI card. System auto-applies the corresponding filter and scrolls to the data table.

KPI 카드 클릭 시 해당 필터 자동 적용 + 데이터 표로 스크롤.

UI Status filter set, table re-rendered, deep-link URL: `?status=below_min`.

4

SYSTEM `cron` · `00:00` daily snapshot

Daily 00:00 KST job writes a full inventory snapshot to `wh_inventory_snapshot` for Δ calculation and audit.

매일 00:00 KST 배치가 전체 재고 스냅샷을 `wh_inventory_snapshot`에 기록 (전일 대비 계산용).

EFFECT Δ column populated next day; data retained 365 days.

5

SYSTEM `trigger` · min-stock alert

On every inventory commit, system checks if `on_hand` crossed below `min_qty` threshold. If so, generates alert + email to warehouse manager and procurement team.

재고 커밋 시마다 최소재고 임계 통과 여부 확인. 미달 시 창고관리자·구매팀에 알림+이메일 발송.

SIDE-EFFECT INSERT `sys_alert`, dedupe within 24h to prevent spam.

✓

USER `click` · `row` → `location`

Clicking a Location cell navigates to WH-04 Location Map centered on the clicked location, with the item highlighted.

위치 셀 클릭 시 WH-04 로케이션 맵으로 이동(해당 위치 중심·품목 하이라이트).

NAV `/wh/location-map?focus=WH-FAB-D-01-1&item=FAB-GY-C03`.

05 Exception & Error Handling

예외 · 에러 처리

E-WH03-001

WebSocket Disconnected

WebSocket 단절

CAUSE	Browser idle > 5min, network issue, or server restart. 유희·네트워크·서버 재시작.
DETECT	No heartbeat received in 30 seconds.
ACTION	Yellow badge in toolbar: "Live updates paused." Auto-reconnect with exponential backoff (2s, 4s, 8s, 16s...).
RECOVERY	On reconnect, full reload triggered to sync any missed updates.

⚠ LIVE PAUSED — RECONNECTING

E-WH03-002

Snapshot Missing

스냅샷 없음

CAUSE	Yesterday's snapshot job failed or DB cleanup removed data.
-------	---

배치 실패·데이터 정리.

DETECT	No row in <code>wh_inventory_snapshot</code> for <code>DATE = CURRENT_DATE - 1</code> .
ACTION	Δ column shows "—" with tooltip: "No snapshot available". KPIs and other columns work normally.
RECOVERY	SYS admin re-runs snapshot job manually from SYS-05 batch monitor.

⚠ NO SNAPSHOT — Δ UNAVAILABLE

E-WH03-003 Excel Export Overflow

엑셀 출력 초과

CAUSE	Filtered result > 10,000 rows. 결과가 1만 행 초과.
DETECT	Server-side count check before file generation.
ACTION	Reject. Modal: "Result exceeds 10,000 rows. Narrow filter or use RPT-02 inventory report."
RECOVERY	Apply more specific filter or use Reports module.

⚠ TOO MANY ROWS — NARROW FILTER

E-WH03-004 No Min/Max Defined

최소·최대 미설정

CAUSE	Item has no row in <code>md_minmax</code> (master data incomplete). 기준정보 미완성.
DETECT	LEFT JOIN returns NULL for <code>min_qty</code> .
ACTION	Status column shows gray "—" badge with tooltip: "Min/Max not set. Configure in MD-09."
RECOVERY	Master Data manager defines min/max in MD-09 → item appears in Below Min / Low Stock KPIs.

ⓘ MIN/MAX NOT SET — SEE MD-09

E-WH03-005 Negative Inventory

음수 재고

CAUSE	Race condition or data anomaly causes <code>on_hand_qty < 0</code> . 동시성·데이터 이상.
DETECT	Row-level check on load. Should never occur (BR-WH-05 enforces).
ACTION	Highlight row with red border + warning icon. Auto-create incident ticket to SYS team. Send email to WH manager.
RECOVERY	WH-05 inventory adjustment with reason code <code>SYSTEM_ANOMALY</code> + supervisor approval.

⚠ NEGATIVE STOCK — INCIDENT CREATED

Empty Result

E-WH03-006 조회 결과 없음

CAUSE Filter combination yields zero items.
필터 조합으로 결과 없음.

DETECT API returns empty array.

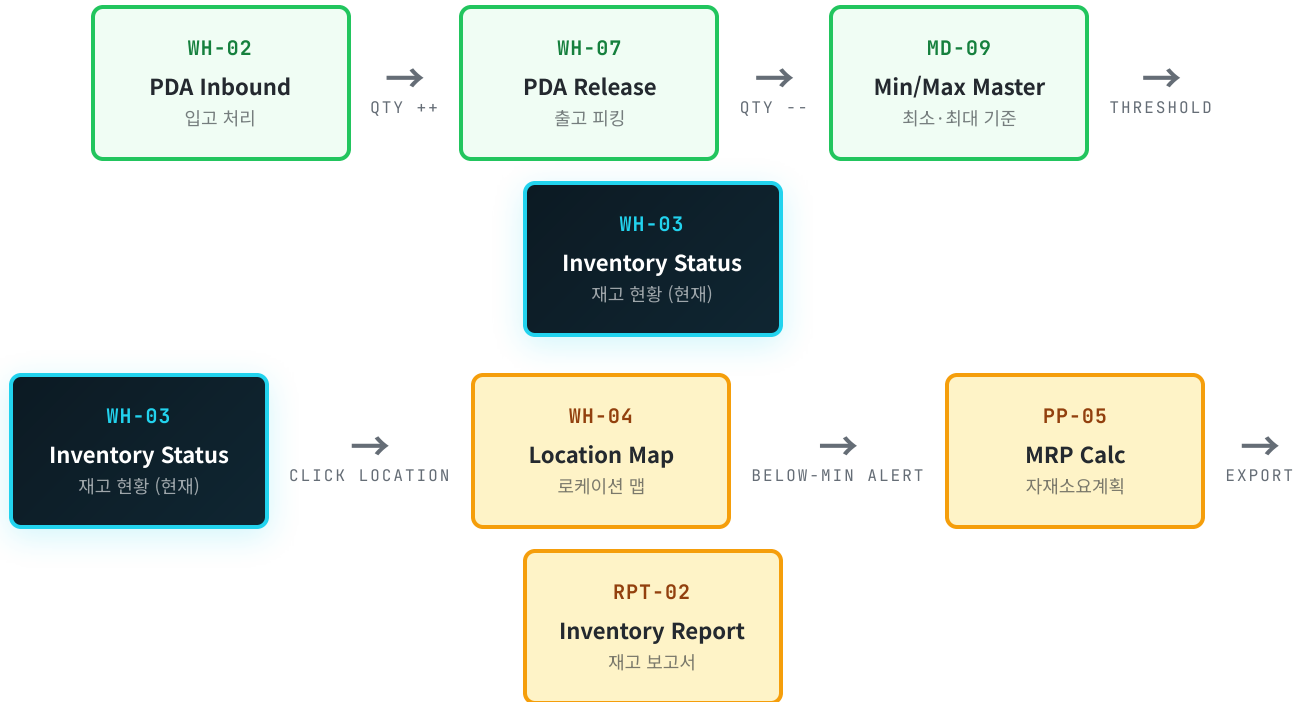
ACTION Empty state illustration + "No items match your filter." Reset Filter button.

RECOVERY User adjusts or resets filters.

! NO RESULTS – ADJUST FILTER

06 Linked Screens

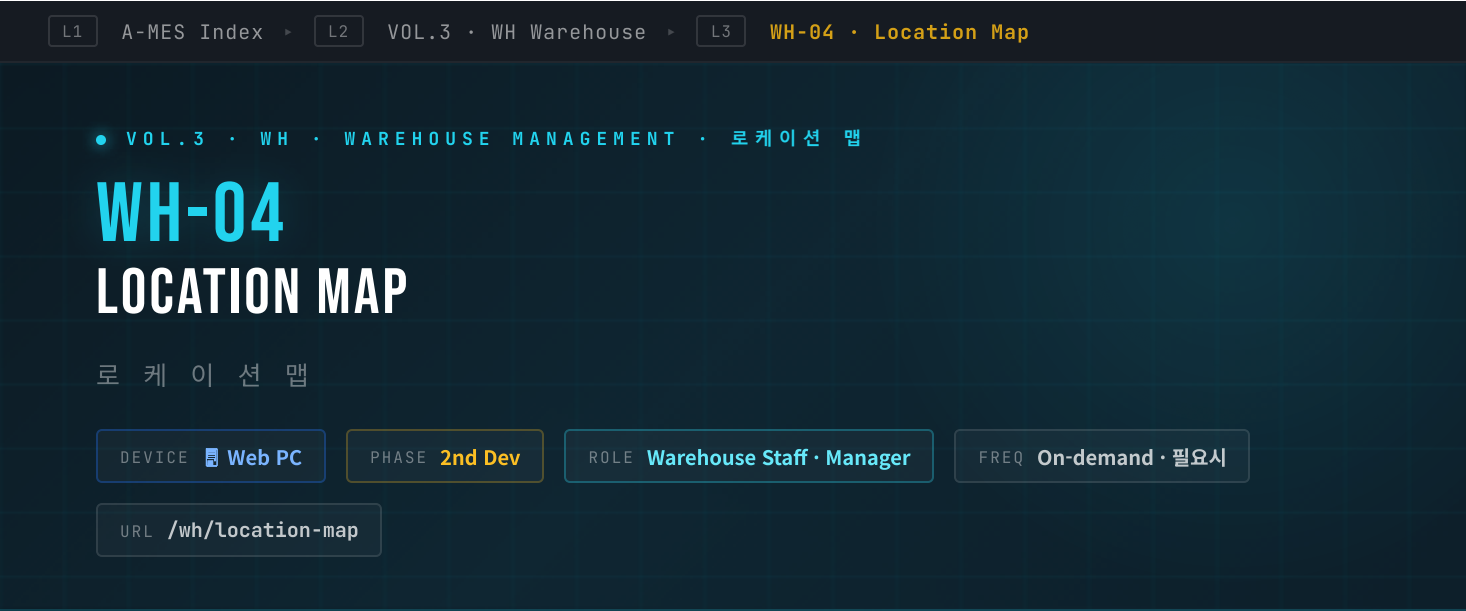
선행 · 후행 연계 화면



07 Applicable Business Rules

적용 업무 규칙

RULE ID	DESCRIPTION	ACTION
BR-WH-03	When on-hand qty falls below the minimum stock level (MD-09), an alert is sent to the warehouse manager and procurement team within 5 minutes. 현재고가 최소재고(MD-09) 미만이 되면 5분 이내 창고관리자·구매팀에 알림 발송.	Alert
BR-WH-05	Inventory quantity must never be negative. Any operation that would result in negative on-hand is rejected at DB constraint level. 재고 수량은 음수가 될 수 없으며, 음수 발생 작업은 DB 제약 조건으로 거부.	Block
BR-WH-06	Daily inventory snapshot at 00:00 KST is mandatory and retained for 365 days for audit and trend analysis. 매일 00:00 KST 재고 스냅샷 필수 + 감사·추이 분석용 365일 보관.	Auto



01 Overview

화면 개요

Visual grid map of all warehouse zones and shelf locations. Each cell represents one storage cell and is color-coded by fill rate: empty (grey), partial (yellow), full (green), or over-capacity (red). Clicking a cell opens a detail panel with stored items, quantities, inbound dates, and a link to the cell's stock movement history. Location codes follow the convention [Warehouse]-[Zone]-[Row]-[Col]-[Level] .

모든 창고 구역과 선반 로케이션을 그리드 형태로 시각화한 화면입니다. 각 셀은 한 개의 저장 위치를 나타내며 충전율에 따라 색상이 다릅니다: 비어있음 (회색), 일부 사용(노랑), 가득 참(초록), 초과(빨강). 셀 클릭 시 보관 품목·수량·입고일·이력 링크가 포함된 상세 패널이 열립니다. 로케이션 코드 체계: [창고]-[구역]-[행]-[열]-[층] .

SCREEN ID	WH-04
LAYOUT	Multi-zone visual grid + side panel 다중 구역 그리드
REFRESH	Real-time (WebSocket) 실시간
PHASE	2nd Dev 2차 개발
CODE FORMAT	WH-FAB-D-01-1
TABLES	md_location , wh_inventory
RBAC	WH.View

02 Screen Mockup

화면 목업 — 시각적 그리드 맵

A-MES · Warehouse · Location Map

Warehouse > Location Map

All Warehouses ▾

All Levels · 전체 층 ▾

Location code search...

Search

Print QR Labels

Excel

240

Total Cells
전체 셀

38

Empty
비어있음

52

Partial
일부 사용

148

Full
가득 참

2

Over Cap
초과

Warehouse Map View · 창고 맵 (Level 1)

☐ Empty · 비어있음 ☐ Partial · 일부 ☐ Full · 가득 ☐ Over · 초과

▼ WH-RM-A · Raw Material Zone A · 원자재 A구역

A-01-1	A-02-1	A-03-1	A-04-1	A-05-1	A-06-1	A-07-1	A-08-1	A-09-1	A-10-1
--------	--------	--------	--------	--------	--------	--------	--------	--------	--------

▼ WH-FAB-D · Fabric Zone D · 원단 D구역 (2 rows × 10 cols)

D-01-1	D-02-1	D-03-1	D-04-1	D-05-1	D-06-1	D-07-1	D-08-1	D-09-1	D-10-1
D-01-2	D-02-2	D-03-2	D-04-2	D-05-2	D-06-2	D-07-2	D-08-2	D-09-2	D-10-2

▼ WH-PNT-E · Paint Zone E · 도료 E구역

E-01-1	E-02-1	E-03-1	E-04-1	E-05-1	E-06-1	E-07-1	E-08-1	E-09-1	E-10-1
--------	--------	--------	--------	--------	--------	--------	--------	--------	--------

Selected Cell · 선택 셀: WH-FAB-D-01-1

Zone Fabric Zone D · Row 1 / Col 1 / Level 1

Capacity 50 rolls

On Hand 42 rolls (84%)

Item FAB-GY-C03

FIFO Date 2025-07-01 09:14

Recent Activity · 최근 거래

IN · 20 rolls · 2025-07-01 09:14 · 김기사

OUT · 5 rolls · 2025-07-02 14:22 · 박작업

IN · 30 rolls · 2025-07-03 11:05 · 김기사

[View full history → WH-08](#)

Fill rate · 충전율 계산 — empty: 0% · partial: 1~80% · full: 80~100% · over: >100% (capacity exceeded). Hover any cell for tooltip with item code and qty.

03

UI Component Specification

UI 컴포넌트 명세

3.1 Filter Toolbar · 필터

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	DEFAULT
CMP-01	Warehouse Filter 창고 필터	SELECT	N	Loaded from <code>md_warehouse</code> . Selecting a warehouse shows only its zones.	All
CMP-02	Level Filter 층 필터	SELECT	N	For multi-level racks. Filters by <code>md_location.level</code> .	L1
CMP-03	Location Search 로케이션 검색	TEXT	N	Searches by <code>location_code</code> or stored <code>item_code</code> . Match → cell flashes + zoom-in.	(empty)
CMP-04	Print QR Labels QR 라벨 출력	BUTTON	N	Opens print modal. Select cells (or zone) → batch-print location QR labels. Requires <code>WH.Manage</code> .	—
CMP-05	Excel Export 엑셀 출력	BUTTON	N	Exports cell inventory list. Includes capacity, on-hand, fill rate %.	—

3.2 Grid Cell Component · 그리드 셀

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	DEFAULT
CMP-06	Cell Color State 셀 색상 상태	DERIVED	—	Auto: 0% → grey · 1–80% → yellow · 80–100% → green · >100% → red.	—
CMP-07	Cell Hover Tooltip 호버 툴팁	TOOLTIP	—	Shows: location_code, item_code, on_hand / capacity, fill %, FIFO date. 200ms delay.	—
CMP-08	Cell Click 셀 클릭	CLICK	—	Opens side panel with full details + activity log. URL updates with <code>?focus=...</code> for deep linking.	—
CMP-09	Empty Cell Action 빈 셀 클릭	CLICK	—	For empty cells, panel shows "Available" state + capacity. Manager can flag as "blocked/maintenance".	—

3.3 Detail Side Panel · 상세 패널

ID	FIELD	SOURCE	FORMAT	EDITABLE
FLD-01	Location Code 위치 코드	md_location.code	Monospace, cyan badge.	No
FLD-02	Zone / Row / Col / Level	md_location	Decomposed display.	No
FLD-03	Capacity 용량	md_location.capacity	Number + unit (rolls/kg/L/EA per item type).	Manager
FLD-04	On Hand / Fill % 현재고·충전율	wh_inventory	Number + percentage. Color matches cell state.	No (use WH-05)
FLD-05	FIFO Date FIFO 기준일	MIN(wh_inbound.inbound_date)	Oldest inbound date in the cell. Datetime format.	No
FLD-06	Recent Activity 최근 거래	wh_transaction	Last 5 entries: IN/OUT, qty, time, operator. Link to WH-08 for full.	No
FLD-07	Location Status 위치 상태	md_location.status	Active / Blocked / Maintenance. Manager can toggle.	Manager

1

USER `page.load` · WH-04 진입

User opens WH-04. System loads location master + current inventory per location. Renders grid color-coded by fill rate. WebSocket subscribes to inventory deltas.

WH-04 진입 시 위치 마스터 + 현재 재고를 로딩하여 충전율 색상 그리드 렌더링. WebSocket 구독.

API `GET /api/wh/locations` · `WS /ws/location.subscribe`.

2

USER `hover` · `cell`

User hovers a cell. Tooltip appears after 200ms with item code, qty, fill %, FIFO date.

셀 호버 시 200ms 지연 후 툴팁 표시(품목·수량·충전율·FIFO 일자).

UI No network call · uses cached data from initial load.

3

USER `click` · `cell`

User clicks a cell. Side panel opens with full detail + recent activity. URL updates with `?focus=WH-FAB-D-01-1` for shareable deep link.

셀 클릭 시 상세 패널이 열리고 최근 거래 표시. URL이 `?focus=...`로 갱신되어 딥링크 공유 가능.

API `GET /api/wh/location/:code/detail`.

4

SYSTEM `push` · `location.updated`

When WH-02 inbound or WH-07 release commits, server pushes the location delta. Cell color and tooltip update with smooth transition animation.

WH-02·WH-07 커밋 시 위치 변경분 푸시. 셀 색상·툴팁이 부드러운 전환과 함께 갱신.

PAYLOAD `{location_code, on_hand, capacity, fill_rate}`.

5

USER `click` · [Print QR Labels]

Manager selects one or more cells (or entire zone) and clicks Print. System generates a printable HTML page with QR codes for each location.

매니저가 셀(또는 구역)을 선택하고 [QR 출력] 클릭 시 QR 코드 포함 인쇄 페이지 생성.

REQUIRE Role `WH.Manage` · QR encodes `location_code` · ZPL or HTML format.

✓

USER `click` · "View history"

From side panel, user clicks "View full history" link. Navigates to WH-08 with location pre-filter applied.

상세 패널에서 [전체 이력] 클릭 시 WH-08로 이동(해당 위치 필터 사전 적용).

NAV `/wh/history?location=WH-FAB-D-01-1`.

05

Exception & Error Handling

예외 · 에러 처리

E-WH04-001

Over-Capacity Detected

초과 적재 감지

CAUSE	Cell's on_hand exceeds defined capacity (data anomaly or manual override). 정의 용량 초과(이상·수동 입력).
DETECT	Cell render: <code>on_hand > capacity</code> .
ACTION	Cell color = red. KPI "Over Cap" increments. Alert to WH manager. Recommend relocation.
RECOVERY	Manager redistributes via WH-05 or expands capacity in MD-11.

🔴 OVER CAPACITY — RELOCATE

E-WH04-002

Location Blocked

위치 차단

CAUSE	Cell marked as <code>Blocked</code> or <code>Maintenance</code> in MD-11. 차단·정비 상태.
DETECT	<code>md_location.status ≠ 'Active'</code> .
ACTION	Cell shown with diagonal stripe pattern. WH-02/WH-07 reject scanning this location until reactivated.
RECOVERY	Manager re-activates location in MD-11.

⚠️ BLOCKED — SEE MD-11

E-WH04-003

Mixed Items in Cell

셀 내 품목 혼재

CAUSE	Multiple item codes stored in same cell (violates BR-WH-07 if not allowed). 한 셀에 복수 품목 보관.
DETECT	<code>COUNT(DISTINCT item_code) > 1</code> per location.
ACTION	Cell badge shows "Mixed". Detail panel lists all items separately. Yellow warning icon.
RECOVERY	If policy disallows, WH-05 adjustment to relocate.

⚠️ MIXED ITEMS — REVIEW

E-WH04-004

QR Print Permission Denied

QR 출력 권한 없음

CAUSE	Non-manager attempts to print QR labels. 매니저 권한 없는 사용자가 출력 시도.
DETECT	Server RBAC check on print request.
ACTION	Return 403. Modal: "QR label printing requires WH.Manage role. Contact your manager."
RECOVERY	Request from manager or escalate.

E-WH04-005

Empty Warehouse View

빈 창고 뷰

CAUSE	Selected warehouse has no defined locations in MD-11. 선택 창고에 위치 미정의.
DETECT	Locations query returns empty.
ACTION	Show illustration: "No locations defined for this warehouse. Configure in MD-11."
RECOVERY	Master data manager defines locations.

NO LOCATIONS — SEE MD-11

E-WH04-006

Rendering Performance

렌더링 성능

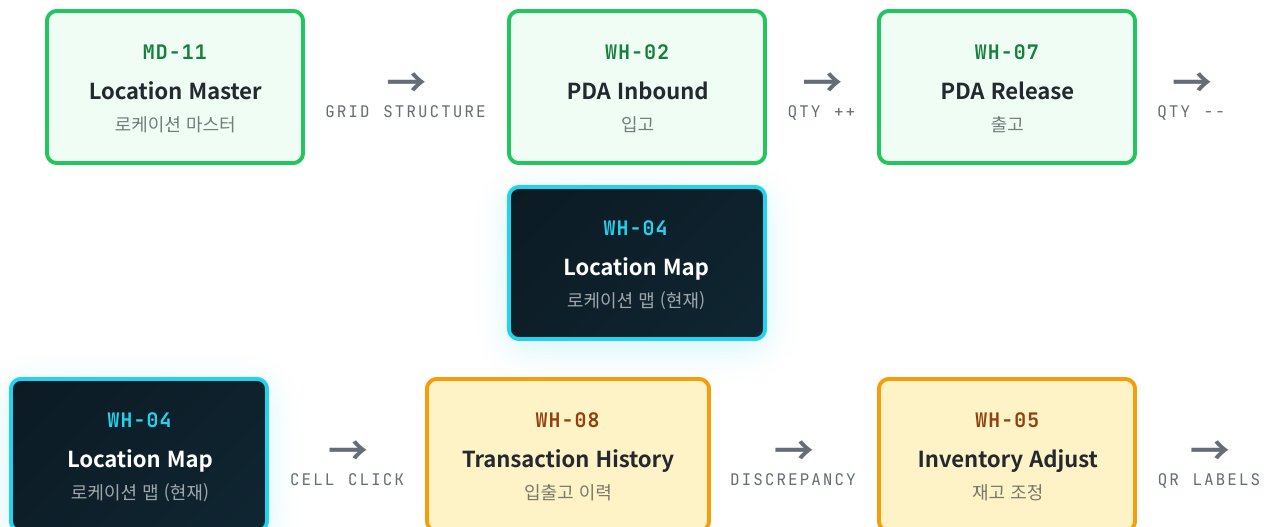
CAUSE	Large warehouse (> 1,000 cells) causes slow render. 대형 창고(1000셀 초과) 렌더링 지연.
DETECT	Cell count check at load time.
ACTION	Activate virtualized rendering (only visible cells). Show progress bar during initial render.
RECOVERY	User applies warehouse/level filter to reduce scope.

LARGE GRID — VIRTUALIZED

06

Linked Screens

선행 · 후행 연계 화면



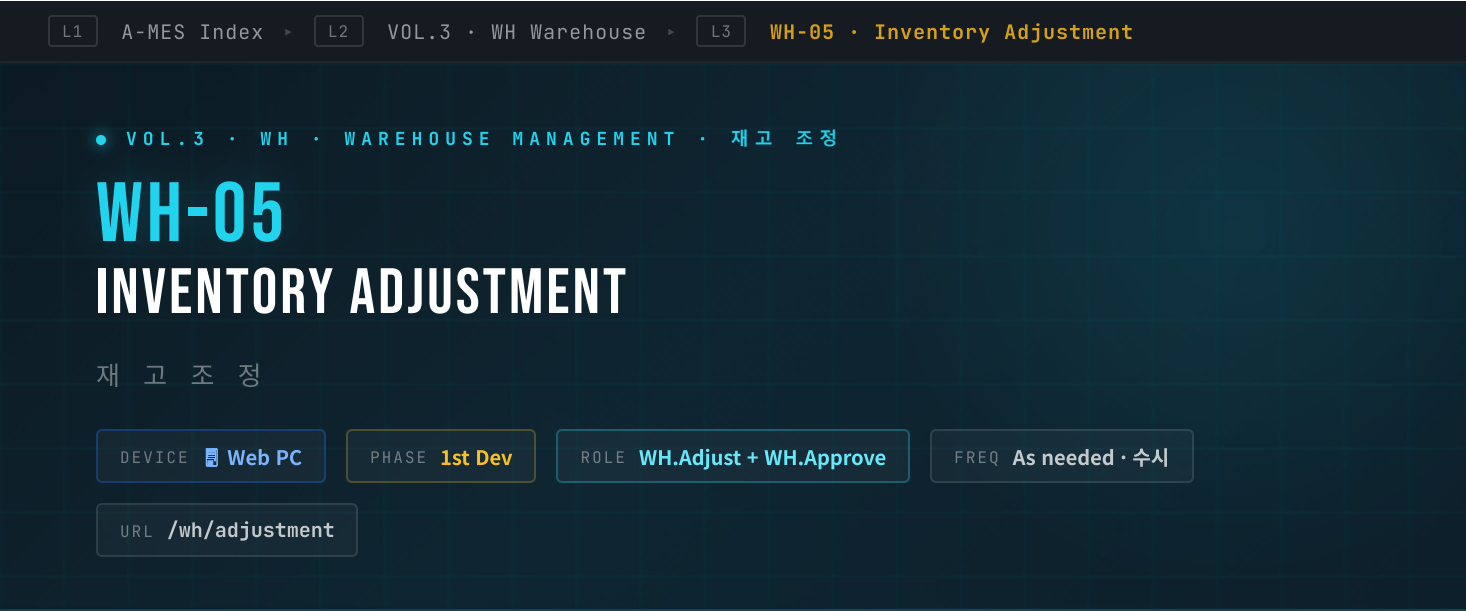
PRINT
Zebra Printer
QR 라벨 출력

07 Applicable Business Rules

적용 업무 규칙

RULE ID	DESCRIPTION	ACTION
BR-WH-07	Each storage cell should hold a single item code. Mixed-item cells require manager approval and are flagged in the map view. 각 셀은 단일 품목 보관 원칙. 혼재 시 매니저 승인 + 맵에 플래그 표시.	Warn
BR-WH-08	Blocked or maintenance-status locations cannot receive new inbound. WH-02 / WH-05 scans are rejected until reactivated. 차단·정비 상태 위치 신규 입고 불가. WH-02·WH-05 스캔 거부.	Block
BR-WH-09	Location capacity is defined in MD-11 and is hard-enforced at WH-02 receiving time. Over-capacity stocking is rejected unless override approved. 로케이션 용량은 MD-11에 정의되며 WH-02 입고 시 강제 적용. 초과 시 거부(승인 시 예외).	Block

A-MES · VOL.3 · WH-04 Location Map · L3 Detail Spec v1.0 · Automotive Parts MES · SAV & GEO Supply ·
Bilingual EN/KO Confidential



01

Overview

화면 개요

Manual correction of system inventory to match physical count results. Every adjustment requires a **reason code** (one of: `COUNT_DIFF` , `DAMAGE` , `EXPIRED` , `SYSTEM_ANOMALY` , `RELOCATION`) and **supervisor approval** before commit. Adjustments above ₩500,000 in value or 10% of on-hand qty automatically escalate to plant manager approval. All adjustments are written to WH-08 transaction history with before/after values, photos (optional), and approver signature.

실사 결과를 시스템 재고에 반영하는 수동 조정 화면입니다. 모든 조정은 **사유 코드**(`COUNT_DIFF` 실사 차이 · `DAMAGE` 파손 · `EXPIRED` 유통기한 · `SYSTEM_ANOMALY` 시스템 이상 · `RELOCATION` 위치 이동)와 **관리자 승인** 후 확정됩니다. 50만원 초과 또는 현재고 10% 초과 조정은 공장장 승인으로 자동 에스컬레이션. 모든 조정은 전/후 값, 사진(선택), 승인자 서명과 함께 WH-08에 기록됩니다.

SCREEN ID	WH-05
LAYOUT	Search + Adjustment form + Approval queue 검색·조정폼·승인대기
APPROVAL	2-level (Supervisor + Plant Mgr) 2단계 승인
THRESHOLD	₩500K or 10% qty → escalate 자동 에스컬레이션
TABLES	wh_adjustment , wh_inventory , wh_transaction
RBAC	WH.Adjust · approver: WH.Approve
PHASE	1st Dev

02

Screen Mockup

화면 목업 — 조정 폼 + 승인 흐름

+ New Adjustment · 신규 조정

All Status

Item / Location / Adj.ID...

Search

Excel

4

Pending Approval
승인 대기

1

Escalated
에스컬레이션

38

Approved · 30d
30일 승인

₩ 2.4M

Adj. Value · 30d
30일 조정금액

NEW ADJUSTMENT FORM · 신규 조정 입력

Item Code *

FAB-GY-C03

Location *

WH-FAB-D-01-1

Current Qty

42 rolls

Counted Qty *

40

rolls

Δ Adjustment

-2 rolls (-4.8%)

Value Impact

-₩ 86,000

Reason Code *

COUNT_DIFF · 실사 차이

Comment

2025-Q3 실사 결과 2 roll 부족 확인 (재
고팀 김기사)

Photo (선택)

Attach Photo

DAMAGE/EXPIRED 시 필

수

Save Draft

Submit for Approval

PENDING APPROVAL · 승인 대기 목록

ADJ-2025-0715-001

Pending · 대기

FAB-GY-C03 · WH-FAB-D-01-1 · -2 rolls (-₩86,000)

Reason: COUNT_DIFF · Requested by: 김기사 · 15:32

✓ Approve

✗ Reject

View Detail

ADJ-2025-0715-002

Escalated · 에스컬레이션

RM-PP-NAT-20 · WH-RM-A-01-2 · -45 kg (-₩630,000)

Reason: DAMAGE · 지게차 충돌 분진 발생 · Plant Mgr
approval required

Waiting Plant Mgr

View Detail

ADJ-2025-0715-003

Draft · 작성중

PNT-BK-NH578 · WH-PNT-E-01-1 · +3 L

Reason: RELOCATION · Draft by: 박작업

⚠ Auto-escalation rules · 자동 에스컬레이션 — 조정 절대값이 50만원 초과 또는 현재고의 10% 초과인 경우 자동으로 공장장(Plant Manager) 승
인으로 전달됩니다. Supervisor만으로는 승인 불가.

03

UI Component Specification

UI 컴포넌트 명세

3.1 Adjustment Form Fields · 조정 품 입력 필드

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	DEFAULT
CMP-01	Item Code 품목코드	AUTOCOMPLETE	Y	Search from <code>md_item</code> . Min 2 chars. Selecting auto-loads location list with on-hand qty.	(empty)
CMP-02	Location 위치	SELECT	Y	Locations where item has on-hand. If empty cell, requires item type compatibility check.	(empty)
CMP-03	Current Qty 현재 수량	DISPLAY	—	Read-only. From <code>wh_inventory.on_hand_qty</code> . Updates live if changed by other transactions.	—
CMP-04	Counted Qty 실사 수량	NUMBER	Y	Must be ≥ 0 . Decimal per item unit. Cannot equal current qty (otherwise no adjustment needed).	(empty)
CMP-05	Δ Adjustment 조정량	CALCULATED	—	<code>counted - current</code> . Display sign + percentage. Red if negative, green if positive.	—
CMP-06	Value Impact 금액 영향	CALCULATED	—	<code>$\Delta \times md_item.unit_cost$</code> . KRW format. Triggers escalation check on input change.	—
CMP-07	Reason Code 사유 코드	SELECT	Y	Options: <code>COUNT_DIFF</code> / <code>DAMAGE</code> / <code>EXPIRED</code> / <code>SYSTEM_ANOMALY</code> / <code>RELOCATION</code> . DAMAGE/EXPIRED → photo required.	(unselected)
CMP-08	Comment 상세 사유	TEXTAREA	Y	Min 10 chars. Max 500. Free-text describing the reason in detail.	(empty)
CMP-09	Photo Attachment 사진 첨부	FILE	N*	JPEG/PNG, max 5MB, up to 3 files. Required when reason = DAMAGE or EXPIRED.	(none)
CMP-10	Save Draft 임시 저장	BUTTON	N	Saves with status=Draft. Does not commit inventory. Can be edited/resumed later.	—
CMP-11	Submit for Approval 승인 요청	BUTTON	Y	Validates all required fields. Status → Pending. Routes to supervisor (or plant mgr if escalated).	—

3.2 Approval Queue Actions · 승인 대기 액션

ID	COMPONENT	TYPE	ROLE	VALIDATION / BEHAVIOR	RESULT
CMP-12	[✓ Approve] 승인	BUTTON	SUP	Only enabled if not escalated. Requires confirmation dialog. Server commits inventory atomically.	Committed
CMP-13	[X Reject] 반려	BUTTON	SUP	Requires rejection reason. Status → Rejected. Notify requester via in-app + email.	Rejected
CMP-14	[View Detail] 상세 보기	BUTTON	ALL	Opens read-only modal with full form data, photos, history.	—
CMP-15	Plant Mgr Approval 공장장 승인	BUTTON	PM	Only for escalated adjustments. Plant Mgr signs (PIN or biometric on tablet). Audit logged.	Committed

3.3 Reason Code Definitions · 사유 코드 정의

CODE	NAME	DEFINITION	PHOTO REQ.	ESCALATION
COUNT_DIFF	Count Difference 실사 차이	Discrepancy between physical count and system qty without obvious cause. Most common reason.	No	By threshold
DAMAGE	Damaged Stock 파손	Physical damage from handling, dropping, contamination. Photo evidence mandatory.	Yes	Always escalate
EXPIRED	Expired Stock 유통기한 경과	Materials past shelf life (e.g. paint, adhesives). Photo + batch info required.	Yes	By threshold
SYSTEM_ANOMALY	System Anomaly 시스템 이상	Data corruption, race condition, or known bug correction. Requires SYS team incident ticket.	No	Always escalate
RELOCATION	Location Move 위치 이동	Internal transfer between locations (paired entry: + and – in two locations). Same value.	No	No

04 Event & Action Flow

이벤트 · 액션 흐름



1

USER click · [+ New Adjustment]

Warehouse staff opens new adjustment form. Requires WH.Adjust role.

창고 담당자가 신규 조정 폼을 엽니다. WH.Adjust 권한 필요.

UI Empty form rendered · item code autocomplete focused.

2

USER select · item + locationUser selects item and location. System fetches current qty and unit cost. Form recalculates Δ and value impact live as counted qty is entered.품목·위치 선택 시 현재고와 단가 자동 로드. 실사 수량 입력에 따라 Δ ·금액 실시간 계산.**API** GET /api/wh/inventory?item=...&loc=... .

3

USER select · reason code

User selects reason. If DAMAGE/EXPIRED, photo upload becomes required. If SYSTEM_ANOMALY, incident ticket ID field appears.

사유 선택 시 DAMAGE/EXPIRED는 사진 필수, SYSTEM_ANOMALY는 인시던트 티켓 ID 입력 필드 표시.

UI Form fields conditionally enabled/required.

4

USER click · [Submit for Approval]All validations pass. System checks escalation thresholds: if $|value| > W500K$ or $|\Delta/current| > 10\%$, routes to Plant Manager; else to Supervisor.

전체 검증 통과 후 에스컬레이션 임계 확인: 금액 50만원 초과 또는 비율 10% 초과 → 공장장, 그 외 → 슈퍼바이저로 라우팅.

API POST /api/wh/adjustment · status='Pending' · approver_level=1|2 .

5

SYSTEM notify · approver

Approver receives in-app notification + email. Approval queue card appears on their WH-05 view. SLA: 4 business hours for Supervisor, 1 business day for Plant Mgr.

승인자에게 인앱·이메일 알림 + 승인 대기 카드 표시. SLA: 슈퍼바이저 4시간, 공장장 1영업일.

SIDE-EFFECT SLA timer starts · auto-reminder at 75% elapsed.

6

APPROVER click · [✓ Approve]

Approver reviews detail + photos. Clicks Approve. System asks confirmation, then commits inventory change atomically in a transaction.

승인자가 상세·사진 확인 후 [승인] 클릭. 확인 다이얼로그 후 트랜잭션으로 재고 변경 원자적 커밋.

API POST /api/wh/adjustment/:id/approve · BEGIN TRAN; UPDATE wh_inventory; INSERT wh_transaction; COMMIT;

✓

SYSTEM complete · downstream

Inventory updated. WH-08 transaction logged with reason code, approver signature, before/after values. Requester notified. Below-min check re-evaluated.

재고 갱신. WH-08에 사유·승인자 서명·전후 값 기록. 요청자에게 결과 알림. 최소재고 재평가.

05 Exception & Error Handling

예외 · 에러 처리

E-WH05-001 Counted Qty = Current

실사·현재 동일

CAUSE	User enters counted qty equal to current. No adjustment needed. 실사 수량이 현재고와 동일.
DETECT	Form validation on Submit.
ACTION	Block submit. Modal: "No adjustment needed — counted qty equals current. Cancel or correct?"
RECOVERY	User cancels or enters different qty.

 NO Δ — CANCEL OR CORRECT

E-WH05-002 Negative Result Stock

음수 재고 결과

CAUSE	Counted qty < 0, or system on_hand would become negative on commit. 결과 재고가 음수.
DETECT	Server validation; DB CHECK constraint also enforces (BR-WH-05).
ACTION	Block. "Counted qty must be ≥ 0 . Inventory cannot be negative."
RECOVERY	User re-counts physical stock.

 NEGATIVE NOT ALLOWED

E-WH05-003 Photo Required (Missing)

사진 누락

CAUSE	Reason is DAMAGE or EXPIRED but no photo attached. 파손·만료 사유인데 사진 미첨부.
DETECT	Form validation on Submit.
ACTION	Block submit. Highlight photo field red. Inline error: "Photo evidence required for DAMAGE/EXPIRED."
RECOVERY	User attaches photo or changes reason.

 PHOTO REQUIRED

E-WH05-004 Concurrent Modification

동시 수정 충돌

CAUSE	Inventory changed (e.g., WH-02 commit) between form open and submit. 폼 열림과 제출 사이에 재고 변경.
DETECT	Optimistic lock: <code>version</code> column mismatch.
ACTION	Reject submit. Modal: "Inventory changed. Current qty is now X. Re-count or adjust?" Show new value.
RECOVERY	User refreshes data and reconsiders.

⚠ STALE DATA — REFRESH

E-WH05-005

Approval SLA Breached

승인 SLA 초과

CAUSE	Approval pending past SLA (4hr Supervisor / 1day Plant Mgr). SLA 시간 초과.
DETECT	Hourly cron checks <code>pending_since</code> .
ACTION	Auto-escalate to next level (Supervisor → Plant Mgr, Plant Mgr → Plant Director). Email all parties.
RECOVERY	Next-level approver acts on request.

⚠ SLA BREACH — AUTO-ESCALATED

E-WH05-006

Self-Approval Attempt

자체 승인 시도

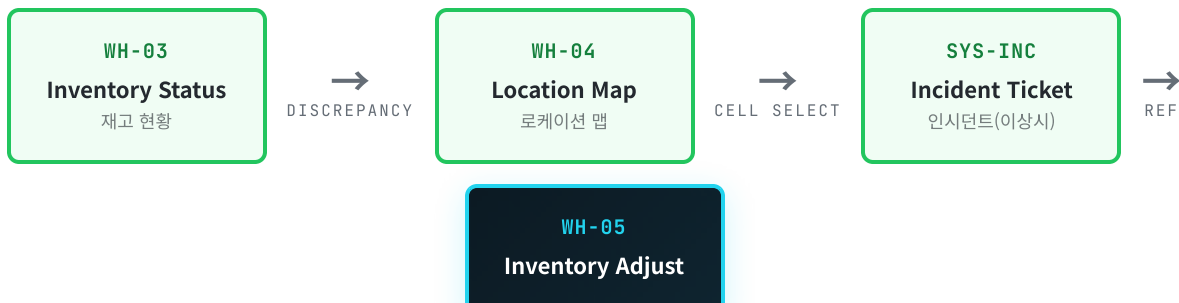
CAUSE	Approver attempts to approve own adjustment request. 요청자가 자신의 요청을 승인 시도.
DETECT	Server check: <code>approver_id = requester_id</code> .
ACTION	Block. "Self-approval is not allowed. Route to another approver." Audit logged.
RECOVERY	Different approver handles request.

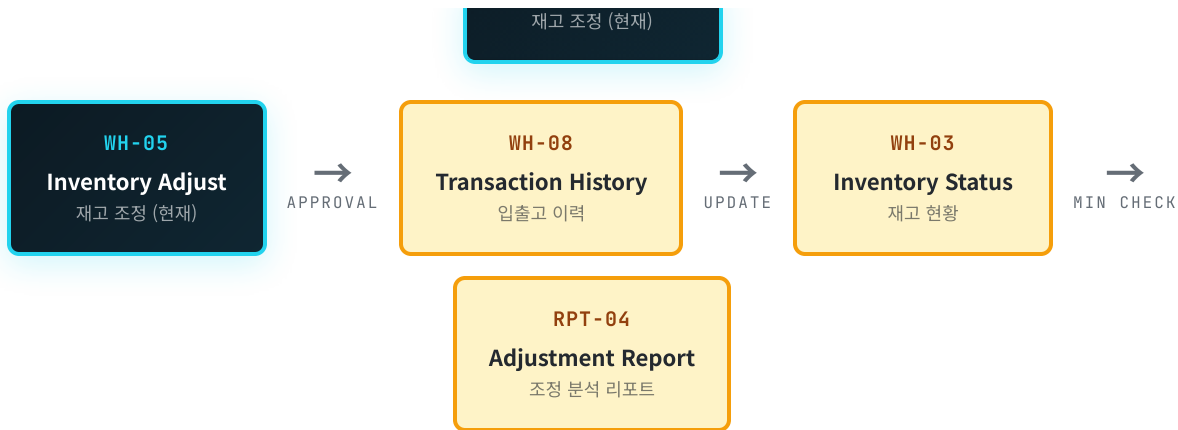
🚫 SELF-APPROVAL FORBIDDEN

06

Linked Screens

선행 · 후행 연계 화면

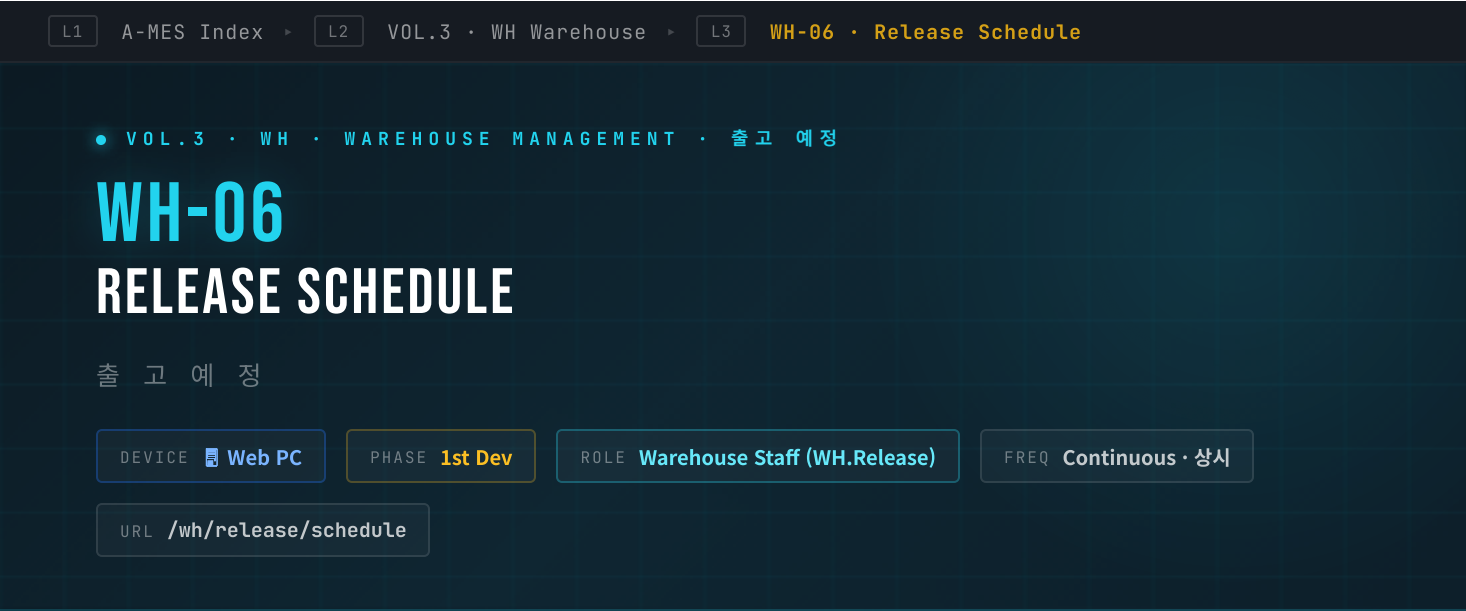




07 Applicable Business Rules

적용 업무 규칙

RULE ID	DESCRIPTION	ACTION
BR-WH-05	Inventory quantity must never be negative. DB-level constraint rejects any adjustment that would result in negative on-hand. 재고는 음수 불가. DB 제약으로 음수 결과 조정 거부.	Block
BR-WH-10	All inventory adjustments require reason code + supervisor approval. Adjustments with value > ₩500K or > 10% of on-hand automatically escalate to plant manager. 모든 재고 조정은 사유코드 + 슈퍼바이저 승인 필수. 50만원 또는 10% 초과 시 공장장 자동 에스컬레이션.	Workflow
BR-WH-11	DAMAGE and EXPIRED reasons require photo evidence (min 1, max 3 photos). Photos retained 5 years for audit. 파손·만료 사유는 사진 증빙 필수(1~3장). 감사용 5년 보관.	Required
BR-WH-12	Self-approval is strictly forbidden. The requester cannot be the approver for the same adjustment. 자체 승인 금지. 요청자와 승인자가 동일인 불가.	Block



01 Overview

화면 개요

Lists upcoming material release demands derived from active Work Orders (WO) in the PP module. For each WO line, the system computes BOM-exploded material needs vs current on-hand and flags shortages. Warehouse staff use this screen to **pre-pick** materials ahead of production start (preparation status), and click **[Start Picking]** to launch WH-07 PDA flow. Filters by line, shift, WO number, and item. ETA is the planned WO start time minus a configurable lead (default: 30 min).

PP 모듈 활성 작업지시(WO) 기반 자재 출고 예정 목록입니다. WO별로 BOM 분해된 자재 소요량을 현재고와 대조하여 부족 자재를 표시합니다. 창고 담당자는 본 화면에서 생산 시작 전 **사전 피킹**을 준비하고(준비 상태), **[Start Picking]** 클릭 시 WH-07 PDA 흐름이 시작됩니다. 라인·교대·WO 번호·품목 필터 지원. ETA는 WO 계획 시작 시각에서 설정된 리드타임(기본 30분)을 차감한 값입니다.

SCREEN ID	WH-06
LAYOUT	Filter + KPI + Demand table 필터·KPI·수요표
SOURCE	PP-04 WO + PP-BOM 작업지시·BOM
REFRESH	Real-time push + 5 min cron 실시간 + 5분
LEAD TIME	30 min (config in MD-15) 기본 30분
TABLES	wo_header , wo_bom_demand , wh_inventory
RBAC	WH.View · WH.Release

02 Screen Mockup

화면 목업

A-MES · Warehouse · Release Schedule

Warehouse > Release > Schedule

All Lines

All Shifts

All Status

연도 - 월 - 일

WO No / Item...

Search

Excel

23

Today's WOs
금일 WO

14

Ready to Pick
피킹 가능

6

Within 30 min
30분 이내 시작

2

△ Shortage
자재 부족

1

Released
출고 완료

WO No. 작업지시	Line · Shift 라인 · 교대	Finished Item 생산 품목	WO Qty	Material to Release 출고 자재	Required 소요량	On Hand 현재고	Plan Start 계획 시작	ETA Lead	Status	Action
WO-2025-0715-001	INJ-L1 · 주간	TRIM-LH-001 트림 LH	2,030	RM-PP-NAT-20	81 kg	1,250 kg	14:00	13:30	Ready	▶ Start Picking
WO-2025-0715-002	IMG-L1 · 주간	TRIM-LH-001-GY 트림 LH 회색	2,030	FAB-GY-C03	12 roll	42 roll	16:00	15:30	Ready	▶ Start Picking
WO-2025-0715-003	PNT-L1 · 주간	CONS-UP-001 콘솔 어퍼	650	PNT-BK-NH578	320 L	284 L △	18:00	17:30	△ Shortage	Insufficient
WO-2025-0715-004	INJ-L2 · 야간	TRIM-RH-001 트림 RH	2,030	RM-PP-NAT-20	81 kg	1,169 kg*	22:00	21:30	Prep	📄 Prepare
WO-2025-0716-001	IMG-L1 · 주간	TRIM-RH-001-GY 트림 RH 회색	2,030	FAB-GY-C03	12 roll	30 roll**	2025-07-16 10:00	D-1	Pending	Detail

* after WO-2025-0715-001 consumes 81 kg (projected) ** after WO-2025-0715-002 consumes 12 roll (projected)

ETA Lead · 리드타임 — Plan Start - Lead (default 30 min). When ETA < CURRENT_TIME, the row turns red and triggers immediate picking alert. Lead time configurable per line in MD-15 Line Master.

03 UI Component Specification

UI 컴포넌트 명세

3.1 Filter Toolbar · 필터

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	DEFAULT
CMP-01	Line Filter 라인 필터	SELECT	N	From md_line active rows. Selecting line filters all downstream lists.	All Lines
CMP-02	Shift Filter 교대 필터	SELECT	N	주간 (Day 08:00–20:00) / 야간 (Night 20:00–08:00).	All Shifts
CMP-03	Status Filter 상태 필터	SELECT	N	Options: Pending / Prep / Ready / Shortage / Released . KPI cards auto-apply.	All Status
CMP-04	Date Picker 날짜 선택	DATE	N	Filter by WO plan start date. Default today.	Today
CMP-05	Search Box 검색창	TEXT	N	Match WO no., item code or item name. Min 2 chars. 300ms debounce.	(empty)
CMP-06	Excel Export 엑셀 출력	BUTTON	N	Filename: WH06_Release_YYYYMMDD.xlsx . Includes shortage column with red highlighting.	—

3.2 KPI Strip · KPI

ID	METRIC	TYPE	CALCULATION	REFRESH
KPI-01	Today's WOs 금일 WO	COUNT	COUNT(*) FROM wo_header WHERE DATE(plan_start) = TODAY AND status IN ('Released', 'Active') .	5 min
KPI-02	Ready to Pick 피킹 가능	COUNT	WOs with all materials on-hand sufficient. Green.	5 min
KPI-03	Within 30 min 30분 이내	COUNT	WOs whose ETA (plan_start - lead) is within next 30 minutes. Yellow urgency.	1 min
KPI-04	⚠ Shortage 자재 부족	COUNT	WOs where any required material's on_hand < required (after upstream allocation). Red.	5 min
KPI-05	Released 출고 완료	COUNT	WOs with all material release transactions complete today.	5 min

3.3 Data Table Columns · 표 컬럼

ID	COLUMN	SOURCE	FORMAT / RULE	SORT
COL-01	WO No. 작업지시	wo_header.wo_no	Monospace. Click → PP-04 WO detail.	ASC
COL-02	Line · Shift 라인 · 교대	wo_header.line_id	Line code + shift name.	—
COL-03	Finished Item 생산 품목	md_item	EN main + KO subtext.	—
COL-04	WO Qty WO 수량	wo_header.plan_qty	Right-align, thousands separator.	—
COL-05	Material 출고 자재	wo_bom_demand.item_code	Multiple rows if multiple materials per WO (one row per material). Monospace.	—
COL-06	Required 소요량	wo_bom_demand.required_qty	WO qty × BOM unit ratio. Right-align with unit.	—
COL-07	On Hand (projected) 현재고 (예상)	Calculated	Current on_hand minus upstream WO allocations. Red if insufficient. Tooltip shows allocation breakdown.	—
COL-08	Plan Start 계획 시작	wo_header.plan_start	Time only if today; full datetime if other day.	ASC ★
COL-09	ETA Lead 출고 ETA	Calculated	plan_start - line.lead_time . Red if past, yellow if within 30min, default else.	—
COL-10	Status 상태	Derived	Pending / Prep / Ready / Shortage / Released. Auto from rules.	—
COL-11	Action 액션	Button	[Start Picking] if Ready · [Prepare] if Prep · [Detail] if Pending · disabled if Shortage.	—

04 Event & Action Flow

이벤트 · 액션 흐름

1

SCHEDULED cron · WO demand refresh @ 5min

Every 5 minutes, system recalculates material demand for all active WOs by joining wo_header × wo_bom_demand × wh_inventory. Updates shortage flags.

5분마다 WO·BOM·재고를 조인하여 자재 소요·부족 플래그 재계산.

EFFECT Materialized view vw_release_schedule refreshed · WebSocket push to clients.

2

USER page.load · WH-06 진입

Warehouse staff opens screen. Default filter: today's date + all lines. Loads demand list sorted by ETA ascending.
WH-06 진입 시 기본 필터(오늘 + 전체 라인)로 ETA 오름차순 정렬 로딩.

API GET /api/wh/release/schedule?date=today .

3

USER cClick · [📄 Prepare]

User marks a WO as "preparing" — soft-reserves materials in inventory (status flag, not yet deducted). Other WOs see projected on-hand updated.

[Prepare] 클릭 시 자재를 소프트 예약(상태만 변경, 차감 아님). 다른 WO는 예상 가용량 갱신 확인.

API POST /api/wh/release/prepare · wo_header.prep_flag=1 · reserved_qty set.

4

USER cClick · [▶ Start Picking]

User initiates picking. System checks RBAC WH.Release and material availability. Creates picking session and launches WH-07 PDA flow.

[Start Picking] 클릭 시 권한 + 자재 가용성 확인 후 피킹 세션 생성, WH-07 PDA 흐름 시작.

REDIRECT POST /api/wh/release/start → returns picking_id → redirect /wh/pda/release?pid=... .

5

SYSTEM alert · ETA approaching

When ETA < 15 min and status is still Pending/Prep, system sends alert to warehouse staff and line supervisor.
ETA 15분 이내인데 Pending/Prep 상태인 경우 창고 담당·라인 슈퍼바이저에 알림 발송.

SIDE-EFFECT In-app push notification · email · sys_alert row inserted.

!

SYSTEM detect · shortage

When required material > projected on_hand, WO turns red. [Start Picking] disabled. Suggestion: trigger MRP rerun or expedite PO from procurement.

소요량 > 예상 현재고 시 WO가 빨간색으로 전환되고 [Start Picking] 비활성. MRP 재계산·긴급 발주 권고.

ALERT WO marked shortage · email to procurement + plant mgr · ref to PP-05 MRP.

✓

SYSTEM return · from WH-07 success

After WH-07 picking completes, WO status = Released. Inventory deducted. Row badge updates. Production line notified that material is staged.

WH-07 피킹 완료 시 WO 상태 = Released. 재고 차감. 생산 라인에 자재 준비 완료 알림.

DOWNSTREAM WH-08 logs · WH-03 push update · PP-04 WO ready signal · INJ/IMG/PNT line dashboards.

05

Exception & Error Handling

예외 · 에러 처리

Material Shortage

E-WH06-001 자재 부족

CAUSE	Required material qty > projected on-hand. 소요량 > 예상 가용량.
DETECT	5-minute cron + real-time on WH-02/05/07 commits.
ACTION	Row red. [Start Picking] disabled. Email procurement + plant mgr. Link to PP-05 MRP rerun.
RECOVERY	Expedite PO via WH-01 [Register Schedule] or transfer stock from another location.

🚫 SHORTAGE — EXPEDITE PROCUREMENT

E-WH06-002

ETA Already Passed

ETA 경과

CAUSE	Current time > ETA but WO not yet picked. 현재 시각이 ETA를 초과했지만 미피킹.
DETECT	Per-row time check on every render.
ACTION	Row blinks red. ETA badge "OVERDUE". Auto-escalate notification every 5 min until picked or cancelled.
RECOVERY	Start picking immediately, or adjust WO plan_start in PP-04.

⚠ OVERDUE — START NOW

E-WH06-003

Concurrent Picking Lock

중복 피킹 시도

CAUSE	Another user has already started picking this WO. 동일 WO를 다른 사용자가 이미 시작.
DETECT	DB row lock on <code>picking_id</code> already exists.
ACTION	Block. Show "Picking already in progress by [user] since [time]. View only?"
RECOVERY	Wait or coordinate with the other user.

🚫 LOCKED — IN PROGRESS

E-WH06-004

WO Cancelled Upstream

WO 상위 취소

CAUSE	WO was cancelled in PP-04 after warehouse already prepared. PP-04에서 WO 취소.
DETECT	WO status change event from PP module.
ACTION	Row removed from list with notification: "WO cancelled. Return prepared materials." Soft-reservation released automatically.
RECOVERY	Return materials to storage. No inventory deduction needed (was only reserved).

⚠ WO CANCELLED — RETURN MATERIALS

E-WH06-005

BOM Not Defined

BOM 미정의

CAUSE	WO created for an item with incomplete BOM in PP module. PP BOM 미정의 품목 WO.
DETECT	LEFT JOIN <code>wo_bom_demand</code> returns 0 rows.
ACTION	Material column shows "—". Row marked as "BOM ERROR" with link to MD-03 BOM Master.
RECOVERY	Master data team defines/completes BOM, then WH-06 row updates.

⚠ NO BOM — SEE MD-03

E-WH06-006

No Picker Assigned

피커 미지정

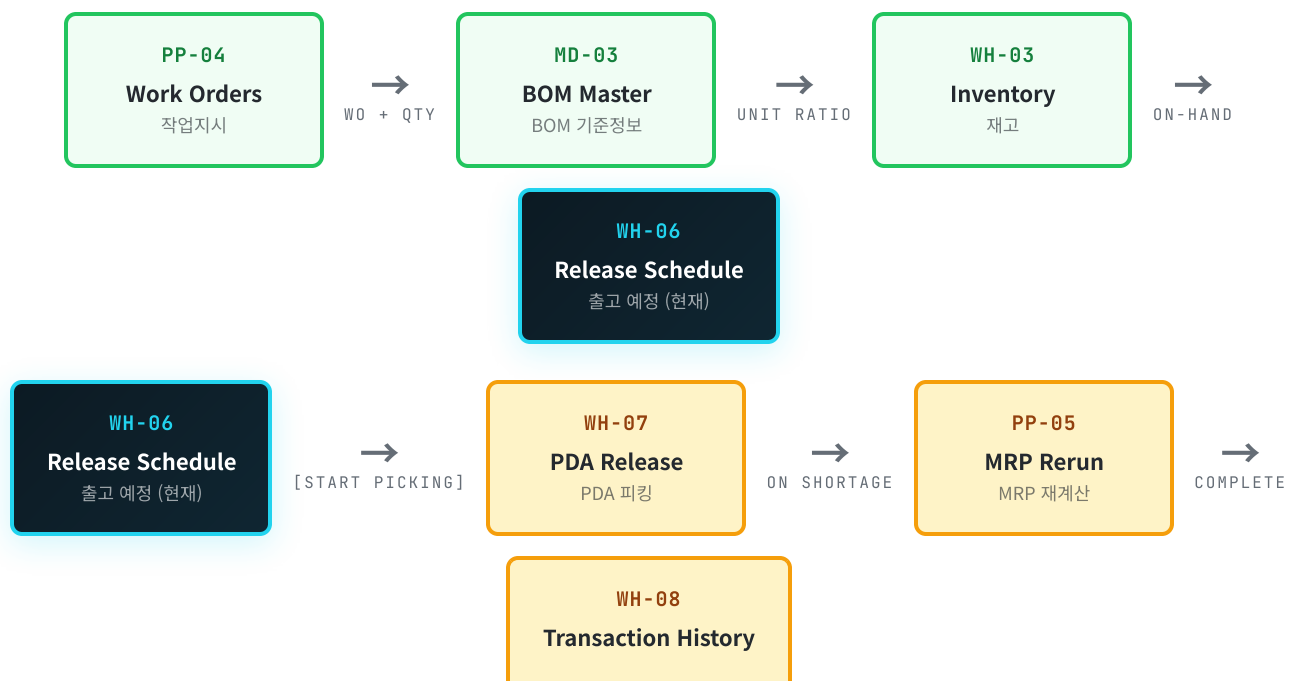
CAUSE	No warehouse staff with <code>WH.Release</code> role is online during shift. 교대 중 출고 권한 담당자 없음.
DETECT	Hourly check during active shifts.
ACTION	Banner: "No picker available for [shift]." Alert sent to WH manager to assign or escalate.
RECOVERY	Manager assigns staff or covers personally.

i NO PICKER — ASSIGN STAFF

06

Linked Screens

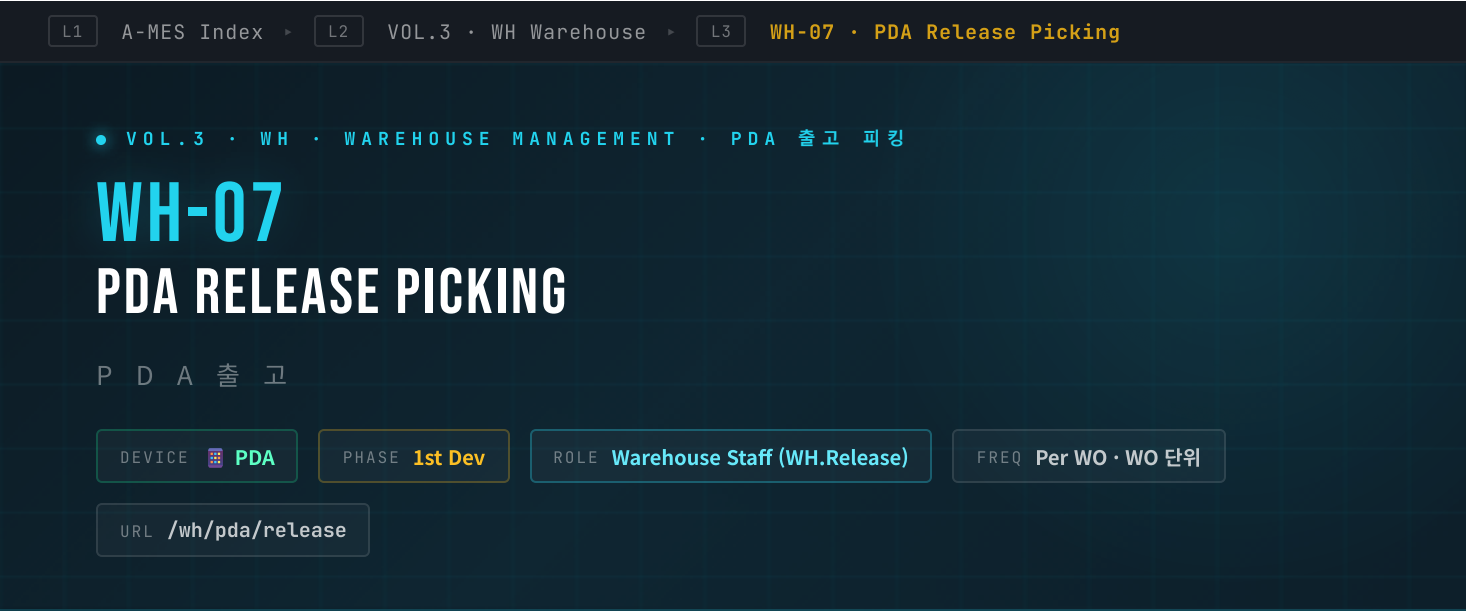
선행 · 후행 연계 화면



07 Applicable Business Rules

적용 업무 규칙

RULE ID	DESCRIPTION	ACTION
BR-WH-04	All outbound picking must follow FIFO (First In First Out) based on inbound date. WH-07 auto-suggests the oldest stock location. 출고 피킹은 입고일 기준 FIFO 자동 적용. WH-07이 최고령 재고 위치 자동 제안.	Auto
BR-WH-13	Picking cannot start if projected on-hand < required after upstream WO allocations. [Start Picking] button is disabled until materials are sufficient. 상위 WO 할당 후 예상 가용량이 소요량 미달이면 피킹 시작 불가. 자재 확보 시까지 버튼 비활성.	Block
BR-WH-14	Concurrent picking on the same WO is blocked by row lock. Only one active picking session per WO at any time. 동일 WO 동시 피킹 차단(행 잠금). 한 WO에 동시에 한 피킹 세션만 활성.	Block



01 Overview

화면 개요

PDA-guided FIFO picking workflow for material release to production lines. Three sequential steps: **(1) Scan WO barcode** to load picking list with system-suggested FIFO locations (oldest stock first), **(2) Scan item barcode at suggested location** — wrong items are immediately blocked, and **(3) Confirm picked quantity** via numeric keypad. Inventory is deducted only after final confirmation. The system enforces FIFO order strictly — if the worker tries a different location, a warning is shown with the correct location code.

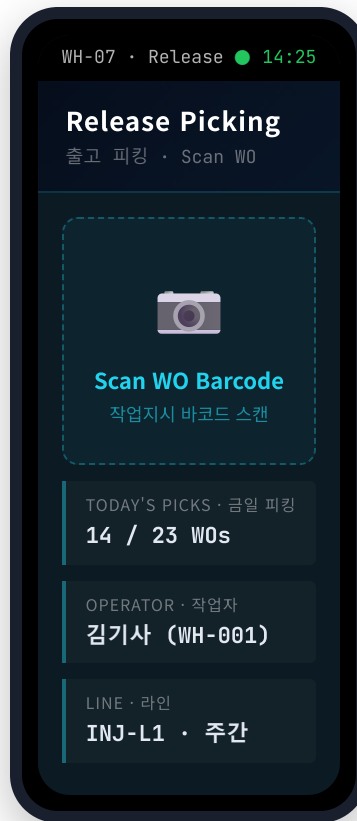
생산 라인 자재 출고를 위한 PDA 기반 FIFO 피킹 흐름입니다. 3단계: **(1) WO 바코드 스캔**으로 시스템이 제안한 FIFO 위치 목록(최고령 재고 우선) 로드, **(2) 제안 위치에서 품목 바코드 스캔** — 오폐품 즉시 차단, **(3) 숫자패드로 피킹 수량 확인**. 최종 확정 후에만 재고 차감. FIFO 순서를 엄격히 강제하며, 다른 위치 시도 시 경고와 함께 올바른 위치 코드를 안내합니다.

SCREEN ID	WH-07
ENTRY	From WH-06 [Start Picking] 출고 예정에서 진입
LAYOUT	3-step PDA wizard 3단계 PDA 위저드
HARDWARE	Zebra TC22 · 1D/2D scanner
RULE	FIFO enforced (BR-WH-04) FIFO 강제
TABLES	wh_release , wh_inventory , wo_bom_demand
RBAC	WH.Release

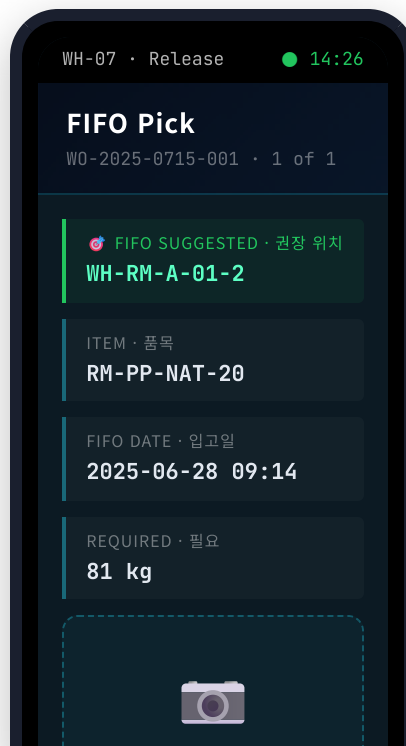
Screen Mockup

● WH-07 · PDA RELEASE PICKING · 3-STEP WIZARD

① SCAN W0 · 작업지시 스캔



② SCAN ITEM @ LOCATION · 위치·품목 스캔



Scan Item Barcode

품목 바코드 스캔

③ CONFIRM QTY · 수량 확정

WH-07 · Release ● 14:27

Confirm Pick Qty

RM-PP-NAT-20 ✓

ITEM VERIFIED ✓
RM-PP-NAT-20

LOCATION · 위치
WH-RM-A-01-2

REQUIRED · 필요
81 kg

81

1 2 3
4 5 6
7 8 9
All 0 ✕

✓ Confirm Release

🚫 Error Case · 에러 케이스 — Wrong Item / Wrong Location · 옳품목·잘못된 위치

WH-07 · Release ● 14:28

🚫 **FIFO VIOLATION**
FIFO 위반 — 피킹 차단

SCANNED · 스캔 위치
WH-RM-A-03-1 ✕

FIFO REQUIRED · FIFO 위치
WH-RM-A-01-2

Trigger · 발생 시점 — Step ② 스캔한 위치가 FIFO 권장 위치와 다른 경우.

Action · 동작 — 즉시 차단 + 빨간 화면 + 진동 + 비프음. 권장 위치 + 입고 일자 차
이 명확히 표시.

AGE DIFFERENCE · 일자 차이
2025-06-28 (older!)

PICK BLOCKED
Scan suggested location · 권장 위치
스캔

Re-scan FIFO Location

Override 옵션 — 슈퍼바이저 PIN 입력 시 1회 예외 허용 (감사 로그 기록, RPT-04에서 통계 추적).

Other errors · 기타 에러 — Wrong item scan, qty > on_hand, location empty, network offline.

03 UI Component Specification

UI 컴포넌트 명세

3.1 Step ① — WO Scan · 작업지시 스캔

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	ON SUCCESS
CMP-01	WO Barcode Scanner WO 바코드 스캐너	SCAN	Y	Reads WO no. Validates against picking session (must match <code>picking_id</code> launched from WH-06).	→ Step ②
CMP-02	Picking List Display 피킹 목록 표시	LIST	—	Shows N of M materials, each with FIFO-suggested location. Ordered by location proximity for efficient route.	—
CMP-03	Manual Entry 수동 입력	BUTTON	N	Fallback if WO label damaged. Keyboard input. Same validation. Logs <code>manual=true</code> .	→ Step ②

3.2 Step ② — Location & Item Scan · 위치 · 품목 스캔

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	ON SUCCESS
CMP-04	FIFO Suggested Display FIFO 권장 위치	LABEL	—	Shows the location with oldest <code>inbound_date</code> for the required item. Green highlight + map preview.	—
CMP-05	Location QR Scanner 위치 QR 스캐너	SCAN	Y	Must match suggested location (BR-WH-04). Override allowed only with supervisor PIN.	→ item scan
CMP-06	Item Barcode Scanner 품목 바코드 스캐너	SCAN	Y	Verifies scanned item matches required <code>item_code</code> . Wrong item → block (E-WH07-002).	→ Step ③
CMP-07	Supervisor Override 슈퍼바이저 예외	PIN	N	6-digit PIN. Allows FIFO violation for one pick. Logged with reason text and timestamp.	→ Step ②

3.3 Step ③ — Quantity Confirmation · 수량 확정

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	ON SUCCESS
CMP-08	Numeric Keypad 숫자 패드	NUMPAD	Y	0-9, decimal. Max 6 digits + 3 decimals. Must be ≤ <code>on_hand</code> at the location.	→ Confirm
CMP-09	[All Required] Button 필요량 전체	BUTTON	N	Auto-fills with required qty from BOM demand. Common case shortcut.	→ Confirm
CMP-10	Qty Display 수량 디스플레이	DISPLAY	—	Large monospace digits. Green if = required, yellow if < required (partial), red if > <code>on_hand</code> .	—
CMP-11	Partial Pick Toggle 부분 피킹	AUTO	—	If qty < required, system asks "Continue to next location for remaining?" and re-routes to next FIFO location.	—
CMP-12	[Confirm Release] 출고 확정	BUTTON	Y	Commits atomically: deduct <code>wh_inventory</code> , insert <code>wh_release</code> , update WO status, log <code>wh_transaction</code> .	Complete

04 Event & Action Flow

이벤트 · 액션 흐름

1

USER launch · from WH-06 [Start Picking]

PDA launches with picking_id context. Loads WO + material list + FIFO-suggested locations.

WH-06 [Start Picking] 클릭으로 PDA 실행. WO·자재 목록·FIFO 권장 위치 로드.

CONTEXT `picking_id`, `wo_no`, materials[] (with FIFO locations).

2

PDA SCAN trigger · WO barcode

Worker scans WO barcode label on the work order travel sheet to confirm picking session.

작업자가 워크오더 종이의 WO 바코드를 스캔하여 피킹 세션 확정.

VALIDATE Scanned WO == session WO · else block.

3

SYSTEM compute · FIFO route

System calculates pick route: for each required material, finds locations with stock and orders by inbound_date ASC (oldest first). Displays first FIFO location.

자재별로 재고 보유 위치를 입고일 오름차순(최고령 우선)으로 정렬하여 피킹 경로 산출. 첫 FIFO 위치 표시.

QUERY `SELECT location, on_hand, MIN(inbound_date) FROM wh_inventory ORDER BY inbound_date ASC`.

4

PDA SCAN trigger · location QR

Worker walks to suggested location and scans the shelf QR code. System validates it matches the FIFO suggestion.

작업자가 권장 위치로 이동 후 선반 QR 스캔. 시스템이 FIFO 일치 여부 검증.

VALIDATE Scanned == suggested · else E-WH07-001 (FIFO violation).

5

PDA SCAN trigger · item barcode

Worker scans the item barcode at the location. System verifies it matches the required item code for this WO.

작업자가 위치의 품목 바코드 스캔. 시스템이 WO 필요 품목과 일치 여부 검증.

VALIDATE `scanned == required_item` · else E-WH07-002 (wrong item).

6

USER tap · numpad / [All Required]

Worker enters picked qty. System checks vs on_hand at location and vs required qty.

수량 입력. 위치 현재고 및 필요량과 대조.

CHECK `qty ≤ location.on_hand` · if < required, partial mode → re-route to next FIFO location.

7

USER tap · [Confirm Release]

Worker confirms. Backend commits transaction atomically: deduct inventory at location, insert release row, log transaction, increment WO picked qty.

확정. 백엔드 트랜잭션 원자적 커밋: 재고 차감, 출고 행 INSERT, 거래 로그, WO 피킹량 증가.

API POST /api/wh/release/confirm · BEGIN TRAN ... COMMIT .



SYSTEM complete · loop or finish

If more materials in picking list, loop to step 4 with next material. When all picked, WO status = Released, line dashboard notified, return to WH-06.

자재가 더 있으면 step 4로 루프. 전부 완료 시 WO Released 처리, 라인 대시보드 알림, WH-06 복귀.

DOWNSTREAM PP-04 WO ready signal · INJ/IMG/PNT line · WH-03/04 update · WH-08 log.

05 Exception & Error Handling

예외 · 에러 처리

E-WH07-001

FIFO Violation

FIFO 위반

CAUSE	Worker scans location with newer stock instead of FIFO-suggested oldest. FIFO 권장 위치가 아닌 다른 위치 스캔.
DETECT	Step ② location scan comparison.
ACTION	Red full-screen + vibration + beep. Show suggested location + age difference. [Re-scan] button.
RECOVERY	Re-scan correct FIFO location · OR supervisor PIN override (audit logged).

🚫 FIFO VIOLATION — SCAN OLDEST

E-WH07-002

Wrong Item Scanned

오폐목 스캔

CAUSE	Scanned item ≠ required item for this WO. 스캔 품목이 WO 필요 품목과 다름.
DETECT	Step ② item scan validation.
ACTION	Block. Red screen + beep + vibration. Required item code shown prominently.
RECOVERY	Re-scan correct item. If physical mislabel suspected, escalate to QC-04.

🚫 WRONG ITEM — RE-SCAN

E-WH07-003

Insufficient at Location

위치 재고 부족

CAUSE	Location on_hand < required (other WO already consumed). 다른 WO가 먼저 소비하여 부족.
DETECT	Step ③ qty entry validation.
ACTION	Allow partial pick of available qty. Auto-route to next FIFO location for remainder.

RECOVERY Continue picking at suggested next location.

⚠ PARTIAL — CONTINUE NEXT LOC

E-WH07-004

Supervisor Override

슈퍼바이저 예외

CAUSE FIFO violation accepted via supervisor PIN.
슈퍼바이저 PIN으로 FIFO 예외.

DETECT Override applied at Step ② re-scan.

ACTION Allow pick to proceed. Mandatory reason text input. Logged with supervisor ID, timestamp, original/actual location.

RECOVERY N/A. Audit reviewed weekly via RPT-04.

⚠ OVERRIDE LOGGED — AUDIT

E-WH07-005

Network Disconnected

네트워크 단절

CAUSE WiFi signal lost during picking.
피킹 중 WiFi 단절.

DETECT Heartbeat timeout 30s.

ACTION Queue confirmation locally (4 hr buffer). Yellow banner. Pick can continue offline; sync on reconnect.

RECOVERY On reconnect, queued releases commit in FIFO order. Conflict resolution if inventory changed meanwhile.

⚠ OFFLINE — QUEUE MODE

E-WH07-006

WO Cancelled During Picking

피킹 중 WO 취소

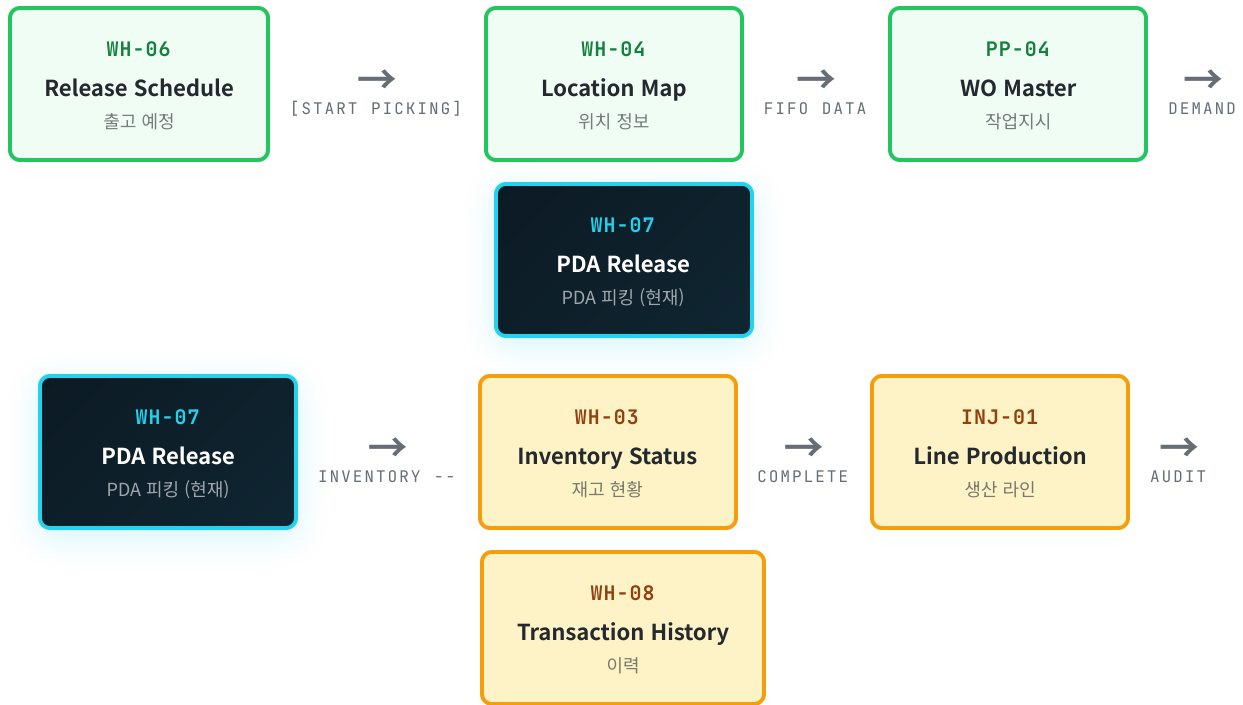
CAUSE PP-04 cancels WO while worker is picking.
피킹 진행 중 WO 취소.

DETECT WO status push event from PP module.

ACTION Pause picking. Modal: "WO cancelled. Return items already picked to original locations?" Auto-create return adjustment ticket.

RECOVERY Worker returns items, system creates compensating release-back transactions.

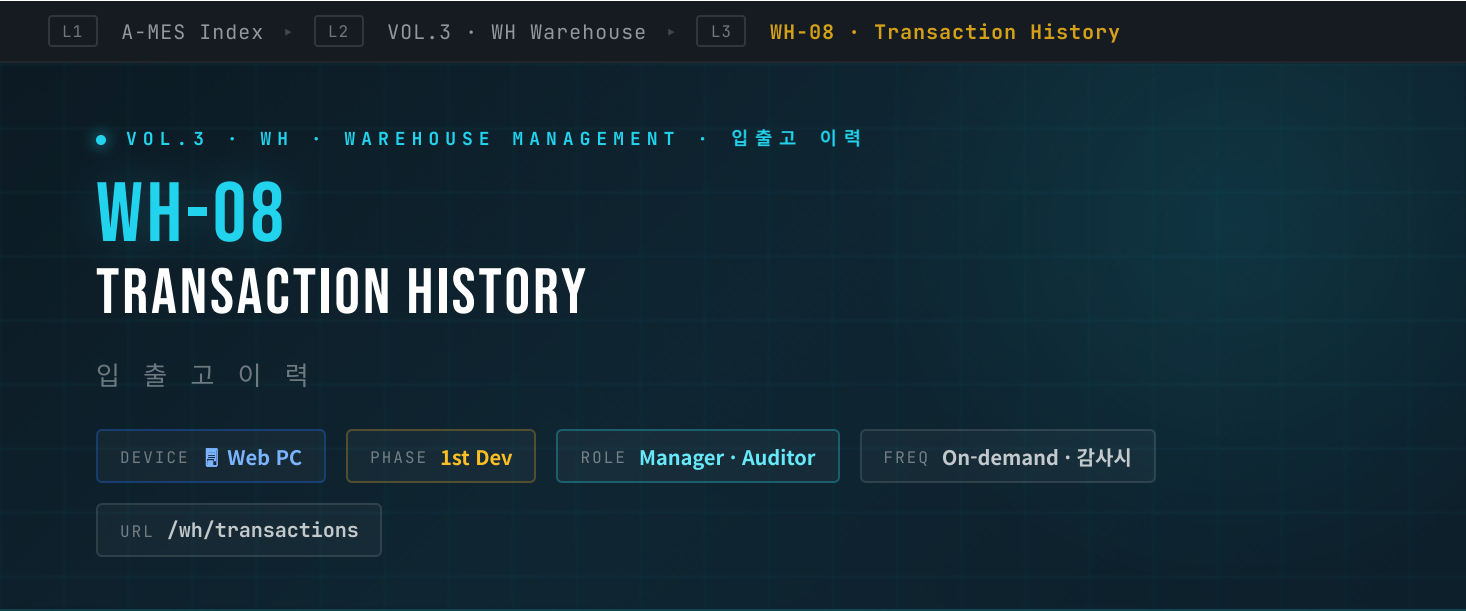
⚠ WO CANCELLED — RETURN ITEMS



07 Applicable Business Rules

적용 업무 규칙

RULE ID	DESCRIPTION	ACTION
BR-WH-04	All outbound picking must follow FIFO based on inbound date. System auto-suggests oldest stock location; deviations require supervisor PIN override with mandatory reason. 출고 피킹은 입고일 기준 FIFO 자동 적용. 권장 위치 외 피킹은 슈퍼바이저 PIN + 사유 입력 필수.	Auto + Override
BR-WH-15	Items scanned during picking must match the required item_code for the WO. Wrong item is immediately blocked with no override option. 피킹 중 스캔 품목은 WO 필요 품목과 일치해야 함. 오폐품 즉시 차단 (예외 없음).	Block
BR-WH-16	FIFO override events are logged and reviewed weekly via RPT-04. More than 5 overrides in 30 days by a single supervisor triggers compliance review. FIFO 예외 이벤트는 RPT-04에서 주간 검토. 30일 내 1인 5건 초과 시 컴플라이언스 리뷰 트리거.	Audit



01 Overview

화면 개요

Complete audit trail of every inventory movement: **IN** (WH-02 inbound), **OUT** (WH-07 release), **ADJ+/ADJ-** (WH-05 adjustment), **RELOC** (location move). Each row captures: timestamp, operator, transaction type, item, location, qty before/after, reason code, reference document (PO / WO / Adjustment ID), and approver (for adjustments). Searchable by date range, item, operator, transaction type, and reference ID. Excel export for audit and regulatory purposes. Records are immutable and retained for **10 years** per quality system requirements.

모든 재고 이동에 대한 완전한 감사 기록: **입고(IN, WH-02)**, **출고(OUT, WH-07)**, **조정(ADJ+/ADJ-, WH-05)**, **위치 이동(RELOC)**. 각 행에는 일시·작업자·거래 유형·품목·위치·전후 수량·사유 코드·참조 문서(PO/WO/Adjustment ID)·승인자(조정의 경우)가 기록됩니다. 날짜 범위·품목·작업자·거래 유형·참조 ID로 검색 가능하며, 감사·규제 목적의 Excel 출력을 지원합니다. 기록은 **수정 불가**하며 품질 시스템 요건에 따라 **10년 보관**됩니다.

SCREEN ID	WH-08
LAYOUT	Filter bar + KPI + Audit table 필터·KPI·감사표
REFRESH	Real-time append (WebSocket) 실시간 추가
RETENTION	10 years · immutable 10년 보관·수정 불가
TABLES	wh_transaction (append-only)
RBAC	WH.Audit · export: WH.Export
PHASE	1st Dev

Screen Mockup

WH-08 · TRANSACTION HISTORY · WEB PC · 1920×1080

A-MES · Warehouse · Transaction History

Warehouse > History

2025-07-15

2025-07-15

All Types

Item / Operator / Ref ID...

All Operators

Search

Excel Export

Audit PDF

28

IN · 입고
금일 입고

42

OUT · 출고
금일 출고

3

ADJ · 조정
금일 조정

1

RELOC · 이동
위치 이동

74

Total Txns
총 거래

Timestamp 일시	Type 유형	Item Code 품목	Location 위치	Qty Before	Δ	Qty After	Reason / Ref 사유 · 참조	Operator 작업자	Approver 승인자
2025-07-15 09:14:23	IN	FAB-GY-C03	WH-FAB-D-01-1	22	+20	42	PO-2025-0701-001 현대원단 발주	김기사	—
2025-07-15 14:27:08	OUT	RM-PP-NAT-20	WH-RM-A-01-2	1,250	-81	1,169	WO-2025-0715-001 INJ-L1 사출	김기사	—
2025-07-15 15:32:11	ADJ-	FAB-GY-C03	WH-FAB-D-01-1	42	-2	40	COUNT_DIFF ADJ-2025-0715-001	김기사	이반장
2025-07-15 16:05:42	IN	PNT-BK-NH578	WH-PNT-E-01-1	234	+50	284	PO-2025-0705-002 KCC도료	박작업	—
2025-07-15 17:42:33	ADJ-	RM-PP-NAT-20	WH-RM-A-01-2	1,169	-45	1,124	DAMAGE ADJ-2025-0715-002 · 지게차 충돌	박작업	공장장
2025-07-15 18:30:17	RELOC	PNT-BK-NH578	WH-PNT-E-01-1 → WH-PNT-E-03-1	284 / 0	±3	281 / 3	RELOCATION ADJ-2025-0715-003	박작업	이반장
2025-07-15 19:14:55	OUT	FAB-GY-C03	WH-FAB-D-01-1	40	-12	28	WO-2025-0715-002 IMG-L1 래핑 · FIFO override (-3일 신상 품)	박작업	이반장 (PIN)

Showing 1-7 of 74 transactions · 1-7 / 74 거래

Prev

1

2

3

...

11

Next

✦ **Immutable audit log · 수정 불가 감사 기록** — 본 화면의 모든 행은 INSERT-only 테이블에 기록되어 수정 · 삭제가 불가능합니다. 변경이 필요할 경우 보정 거래(compensating transaction)를 별도 행으로 기록합니다. 행 색상: 노랑(FIFO Override), 빨강(고가 Damage), 일반.

03 UI Component Specification

UI 컴포넌트 명세

3.1 Filter Toolbar · 필터

ID	COMPONENT	TYPE	REQ	VALIDATION / BEHAVIOR	DEFAULT
CMP-01	Date Range 기간	DATE × 2	Y	From ≤ To. Max range 90 days. Beyond 90 days requires WH.Audit.Extended role.	Today
CMP-02	Type Filter 유형 필터	SELECT	N	Options: ALL / IN / OUT / ADJ / RELOC . Multi-select v2.	All Types
CMP-03	Search Box 검색창	TEXT	N	Search by item code/name, location code, or reference ID (PO/WO/ADJ). 300ms debounce.	(empty)
CMP-04	Operator Filter 작업자 필터	SELECT	N	From md_user where role includes WH.* roles. Searchable dropdown.	All
CMP-05	Excel Export 엑셀 출력	BUTTON	N	Requires WH.Export role. Max 50,000 rows. Filename: WH08_Txn_YYYYMMDD_YYYYMMDD.xlsx .	—
CMP-06	Audit PDF 감사 PDF	BUTTON	N	Generates PDF with cover page (period, filters, generator name) and audit footer on every page. Digitally signed.	—

3.2 KPI Strip · KPI

ID	METRIC	TYPE	CALCULATION	FILTER ON CLICK
KPI-01	IN 입고	COUNT	COUNT WHERE type='IN' within date range.	type=IN
KPI-02	OUT 출고	COUNT	COUNT WHERE type='OUT' .	type=OUT
KPI-03	ADJ 조정	COUNT	COUNT WHERE type IN ('ADJ+', 'ADJ-') .	type=ADJ
KPI-04	RELOC 위치 이동	COUNT	COUNT WHERE type='RELOC' . Paired entries (– and +).	type=RELOC
KPI-05	Total Txns 총 거래	COUNT	Sum of all types.	All

3.3 Table Columns · 표 컬럼

ID	COLUMN	SOURCE	FORMAT / RULE	SORT
COL-01	Timestamp 일시	wh_transaction.txn_time	Format: YYYY-MM-DD HH:MM:SS . KST timezone. Monospace.	DESC ★
COL-02	Type 유형	wh_transaction.type	Badge: IN (green) / OUT (red) / ADJ+/- (yellow/red) / RELOC (purple).	—
COL-03	Item Code 품목	md_item	Monospace. Click → MD-04 item master.	—
COL-04	Location 위치	md_location	For RELOC: shows from → to. Click → WH-04 map.	—
COL-05	Qty Before 전 수량	wh_transaction.qty_before	Right-align. Numerical snapshot at time of transaction.	—
COL-06	Δ 증감	wh_transaction.delta	Signed. Green positive, red negative, purple paired (RELOC).	—
COL-07	Qty After 후 수량	wh_transaction.qty_after	Right-align, bold. = qty_before + delta.	—
COL-08	Reason / Ref 사유·참조	wh_transaction.reason_code + ref_id	Reason code (bold) + ref doc (PO/WO/ADJ ID). Click ref → opens source screen.	—
COL-09	Operator 작업자	md_user.name	User who performed transaction. Click → user profile.	—
COL-10	Approver 승인자	wh_transaction.approver_id	Populated for ADJ/RELOC/Override. "—" for routine IN/OUT.	—

04 Event & Action Flow

이벤트 · 액션 흐름

1

SYSTEM trigger · transaction commit

Whenever WH-02/05/07 commits an inventory change, a corresponding row is written to wh_transaction via DB trigger. Atomic with source operation.

WH-02-05-07이 재고 변경 커밋 시, DB 트리거로 wh_transaction에 행이 자동 기록됨. 원자적 커밋.

EFFECT INSERT INTO wh_transaction (...) VALUES (...) · WebSocket push.

2

USER page.load · WH-08 진입

Auditor or warehouse manager opens screen. Default filter: today, all types. Loads first page of 50 rows sorted by timestamp DESC.

감사자 또는 관리자가 진입. 기본 필터(오늘·전체 유형)로 첫 50행 로딩(시각 내림차순).

API GET /api/wh/transactions?date=today&page=1&size=50 .

3

SYSTEM push · new transaction

When a new transaction commits, WebSocket pushes the row to all subscribed clients. New row appears at top with brief flash animation.

신규 거래 커밋 시 WebSocket으로 푸시. 표 상단에 새 행이 짧은 플래시 애니메이션과 함께 추가.

UI Row prepended · auto-pagination preserved · KPI counters increment.

4

USER cClick · KPI card

User clicks a KPI to drill into that type. Filter applied, table redraws.

KPI 클릭 시 해당 유형 필터 자동 적용 후 표 갱신.

URL ?type=ADJ&date=... · deep-linkable.

5

USER cClick · ref_id link

User clicks a reference (PO / WO / ADJ ID) to navigate to source document.

참조 ID(PO/WO/ADJ) 클릭 시 원본 문서 화면으로 이동.

NAV PO → WH-01 detail · WO → PP-04 · ADJ → WH-05.

6

USER cClick · [📄 Excel Export]

User requests export. System checks WH.Export role and row count. If < 50,000, generates .xlsx async via background job; download link emailed when ready.

Excel 출력 요청. 권한·행 수 검증 후 비동기 백그라운드 작업으로 생성. 완료 시 다운로드 링크 이메일 전송.

API POST /api/wh/export · job_id returned · email when ready.

✓

USER cClick · [📄 Audit PDF]

Generates a digitally signed PDF report with cover page, audit footer, and tamper-evident hash. Used for ISO/IATF audit submission.

디지털 서명된 PDF 감사 보고서 생성(표지·푸터·해시값 포함). ISO/IATF 감사 제출용.

OUTPUT PDF with SHA-256 hash footer · timestamped digital signature.

05 Exception & Error Handling

예외 · 에러 처리

E-WH08-001

Range Exceeds 90 Days

기간 90일 초과

CAUSE Date range > 90 days. Performance and storage concern.
조회 기간 90일 초과.

DETECT Client + server validation.

ACTION	Block. Unless user has <code>WH.Audit.Extended</code> role. Modal: "Range > 90 days requires extended audit role."
RECOVERY	Reduce range or request extended role from SYS admin.

⚠ RANGE TOO LARGE — REDUCE OR ESCALATE

E-WH08-002

Edit Attempt

수정 시도

CAUSE	User attempts to edit/delete a transaction row via API or DevTools. 감사 행 수정·삭제 시도.
DETECT	UPDATE/DELETE on <code>wh_transaction</code> blocked at DB level (BR-WH-17).
ACTION	Return 403. Log attempt with user_id, IP, timestamp in <code>sys_security_audit</code> . Alert SYS admin.
RECOVERY	If correction needed, create compensating transaction via WH-05 with reason <code>SYSTEM_ANOMALY</code> .

🚫 IMMUTABLE — CREATE COMPENSATION

E-WH08-003

Export Row Limit

출력 행 제한

CAUSE	Filtered result > 50,000 rows. 결과 5만 행 초과.
DETECT	Server-side count check.
ACTION	Reject. Suggest narrower filter or use RPT-04 audit reports for large datasets.
RECOVERY	Narrow filter, or request RPT-04 audit batch report.

⚠ TOO MANY ROWS — USE RPT-04

E-WH08-004

Hash Mismatch

해시 불일치

CAUSE	Periodic integrity check detects record hash mismatch (data tampered or corruption). 레코드 해시 불일치(변조·손상).
DETECT	Daily integrity cron at 02:00 KST.
ACTION	CRITICAL alert to SYS admin, plant director, QA lead. Affected rows flagged. Forensic backup loaded.
RECOVERY	SYS team investigates; restore from immutable archive; report to compliance.

🚫 INTEGRITY VIOLATION — INVESTIGATE

E-WH08-005

Old Data Archived

오래된 데이터 아카이브

CAUSE	Searching for data older than 2 years; moved to cold storage.
-------	---

2년 초과 데이터는 콜드 스토리지.

DETECT	Query response time slow (> 5s).
ACTION	Show banner: "Querying archived data. May take longer." Background async fetch + email when ready.
RECOVERY	Wait for fetch or use specific filters to narrow.

i ARCHIVED — FETCHING ASYNC

E-WH08-006 No Transactions Found

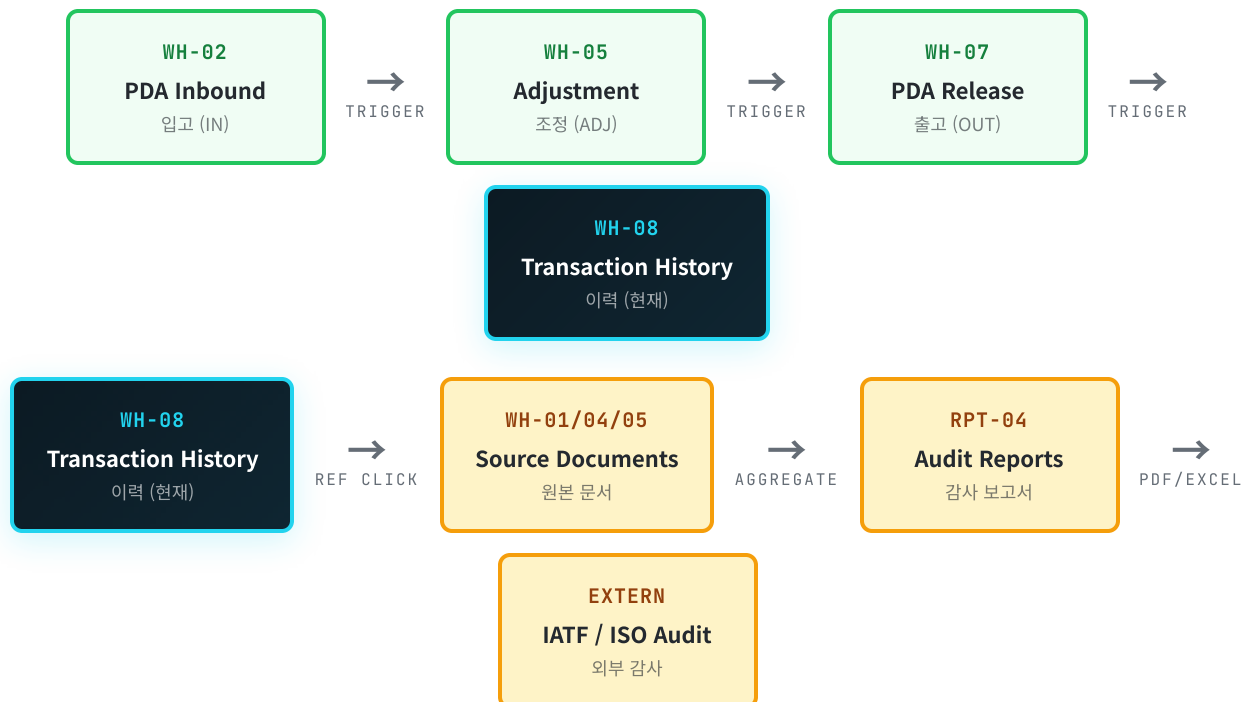
거래 없음

CAUSE	Filter combination yields zero results. 필터 결과 0건.
DETECT	API returns empty array.
ACTION	Empty state illustration: "No transactions match your filters." Reset button.
RECOVERY	Reset or adjust filters.

i NO RESULTS — ADJUST FILTER

06 Linked Screens

선행 · 후행 연계 화면



07

Applicable Business Rules

적용 업무 규칙

RULE ID	DESCRIPTION	ACTION
BR-WH-17	All transactions in wh_transaction are immutable (INSERT-only). UPDATE and DELETE are forbidden at DB level. Corrections must use compensating transactions. 모든 거래는 수정 불가(INSERT 전용). UPDATE·DELETE는 DB 차원에서 금지. 보정은 별도 보정 거래로 처리.	Block
BR-WH-18	Transaction records are retained for 10 years to comply with IATF 16949 traceability requirements. Records older than 2 years move to cold storage. IATF 16949 추적 요건에 따라 거래 기록 10년 보관. 2년 초과 데이터는 콜드 스토리지 이관.	Auto
BR-WH-19	Each transaction row includes a SHA-256 hash of its content. Daily integrity check verifies no tampering. Hash mismatch = CRITICAL incident. 각 거래 행에 SHA-256 해시 포함. 일일 무결성 검증으로 변조 감지. 불일치는 치명적 인시던트.	Integrity